Integrating Enhanced RFM-Based Feature Engineering and Selection for Credit Card Fraud Detection

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Outline

1. Motivation and Introduction
2. Data
3. Methodology
4. Study plan
5. Results
6. Conclusion

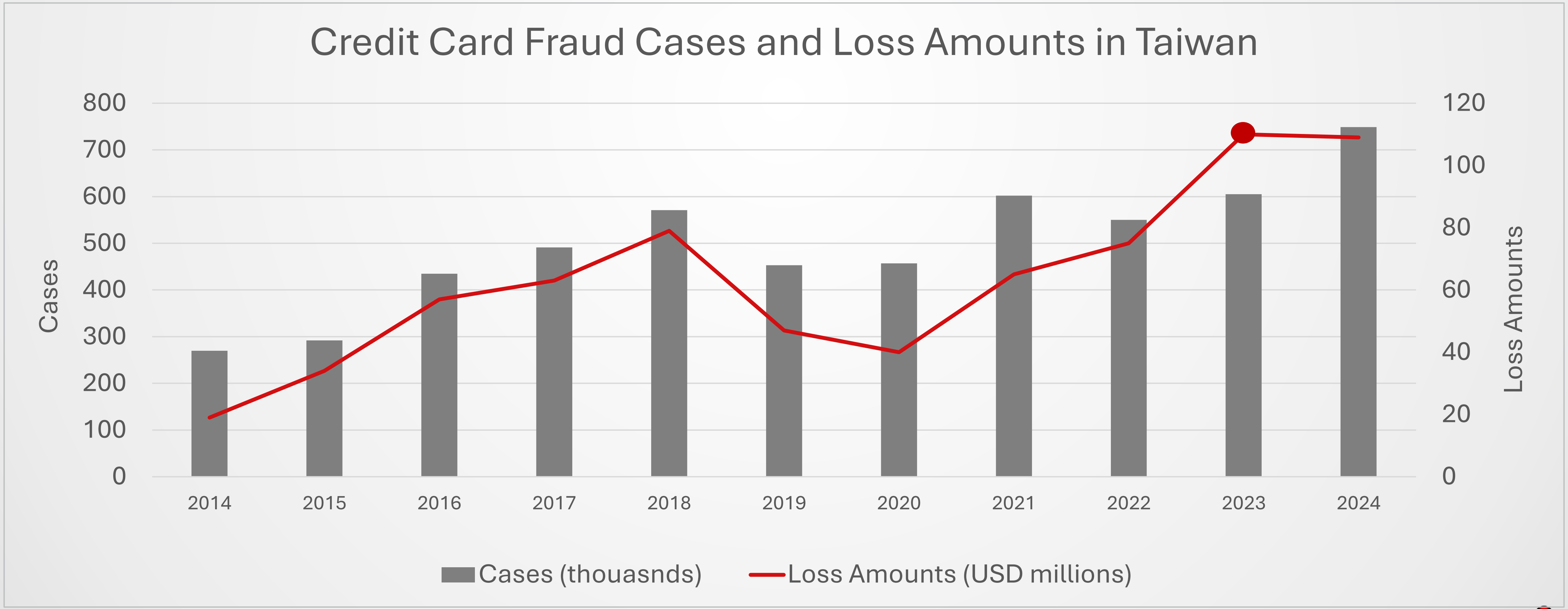
Importance of Credit Card Fraud Detection

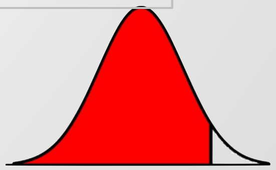
World: Fraud losses reached US$ 33.83 billion in 2023 and US$33.45 billion in 2022.

United States: Losing over US$10 billion to fraud in 2023 with a 14% increase from 2022.

Europe: Fraud totaled US$4.9 billion in 2022.

Taiwan: Fraud losses hit US$110 million in 2023, the highest in a decade.





Characteristics of Fraud Detection

Concealed

Carefully planned and hard to capture by fixed rules.

Evolving

Continuously changing, requiring model adaptability.

Rare

Low occurrence causes class imbalance and weak learning.

(Bart Baesens et al., 2021)

Feature engineering frameworks in fraud detection

1. RFM framework

Component	What It Measures	Fraud Detection Insight
Recency	How recently a customer performed a specific type of transaction.	Long gaps in expected behavior may indicate suspicious activity.
Frequency	How often a customer makes certain types of transactions within a time window.	Sudden surge or drop to zero may signal abnormal or fraudulent activity.
Monetary	The total or average amount of money spent in transactions over a given period.	Unusually high amounts can be a strong fraud indicator.

2. Transaction Aggregation in HOBA

Homogeneity-Oriented Behavior Analysis (HOBA) is an improved RFM framework.

Aggregation Characteristic	Aggregation Period	Transaction Behavior Measure	Aggregation Statistics
• purchase • abnormal time • foreign transaction country • high-risk MCC • online transaction	• past 1,3,5,7 day/ transaction	• transaction amount • transaction time gap • geographical distance interval • open-to-buy utilization rate	• count • mean • sum • standard deviation

(Bart Baesens et al., 2021) (Xinwei Zhang et al., 2019)

Financial Transactions Dataset: Analytics

Dataset Overview

- Source: Kaggle
- Duration: 2010/01/01-2019/10/31
- Size: 13,305,915 Observations (Only 8,914,963 have labels)

Composition

- Source datasets (5): Transaction, User, Credit Card, Train_Fraud_Labels, and MCC_Codes.
- Columns: 37 (expanded to 51 after one-hot encoding and missing value imputation)
- Target labels: fraud_label
- Overall Fraud Rate: 0.100 (0.149% among labeled samples)

Source columns

Red: join Key | Green: missing_flag | Blue: one hot encoding

users_df	
0	client_id
1	current_age
2	retirement_age
3	birth_year
4	birth_month
5	gender
6	address
7	latitude
8	longitude
9	per_capita_income
10	yearly_income
11	total_debt
12	credit_score
13	num_credit_cards

mcc_codes_df	
0	mcc_code
1	category_name

train_fraud_labels_df	
0	transaction_id
1	fraud_label
2	fraud_label_missing_flag

transactions_df	
0	transaction_id
1	Date
2	client_id
3	card_id
4	amount
5	use_chip
6	merchant_id
7	merchant_city
8	merchant_state
9	zip
10	mcc_code
11	errors
12	merchant_state_missing_flag
13	zip_missing_flag
14	errors_missing_flag
15	use_chip_Chip Transaction
16	use_chip_Online Transaction
17	use_chip_Swipe Transaction

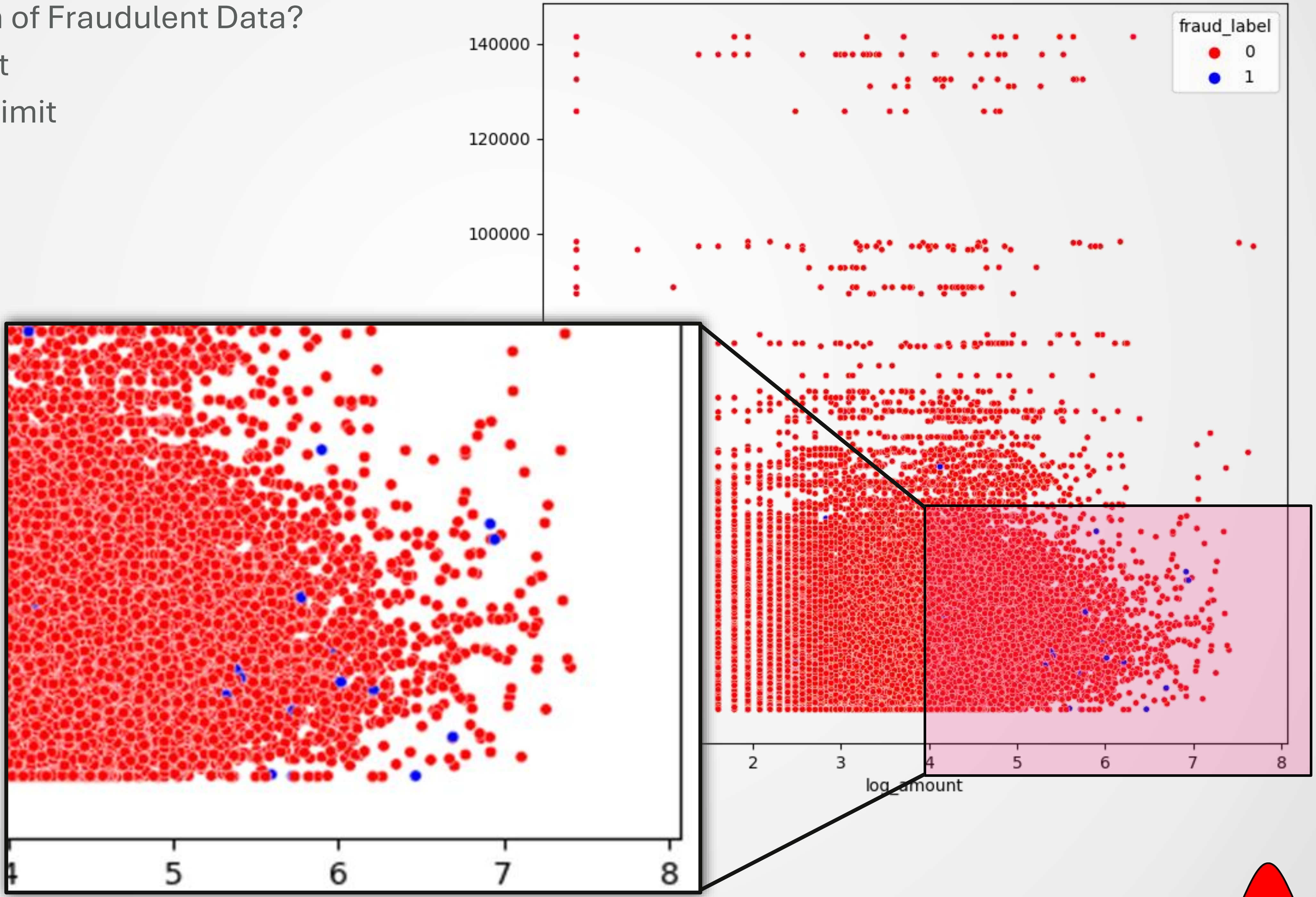
cards_df	
0	card_id
1	client_id
2	card_brand
3	card_type
4	card_number
5	expires
6	cvv
7	has_chip
8	num_cards_issued
9	credit_limit
10	acct_open_date
11	year_pin_last_changed
12	card_on_dark_web
13	card_type_Credit
14	card_type_Debit
16	card_brand_Amex
17	card_brand_Discover
18	card_brand_Mastercard
19	card_brand_Visa

Correlation–Key Finding in Scatter Matrix

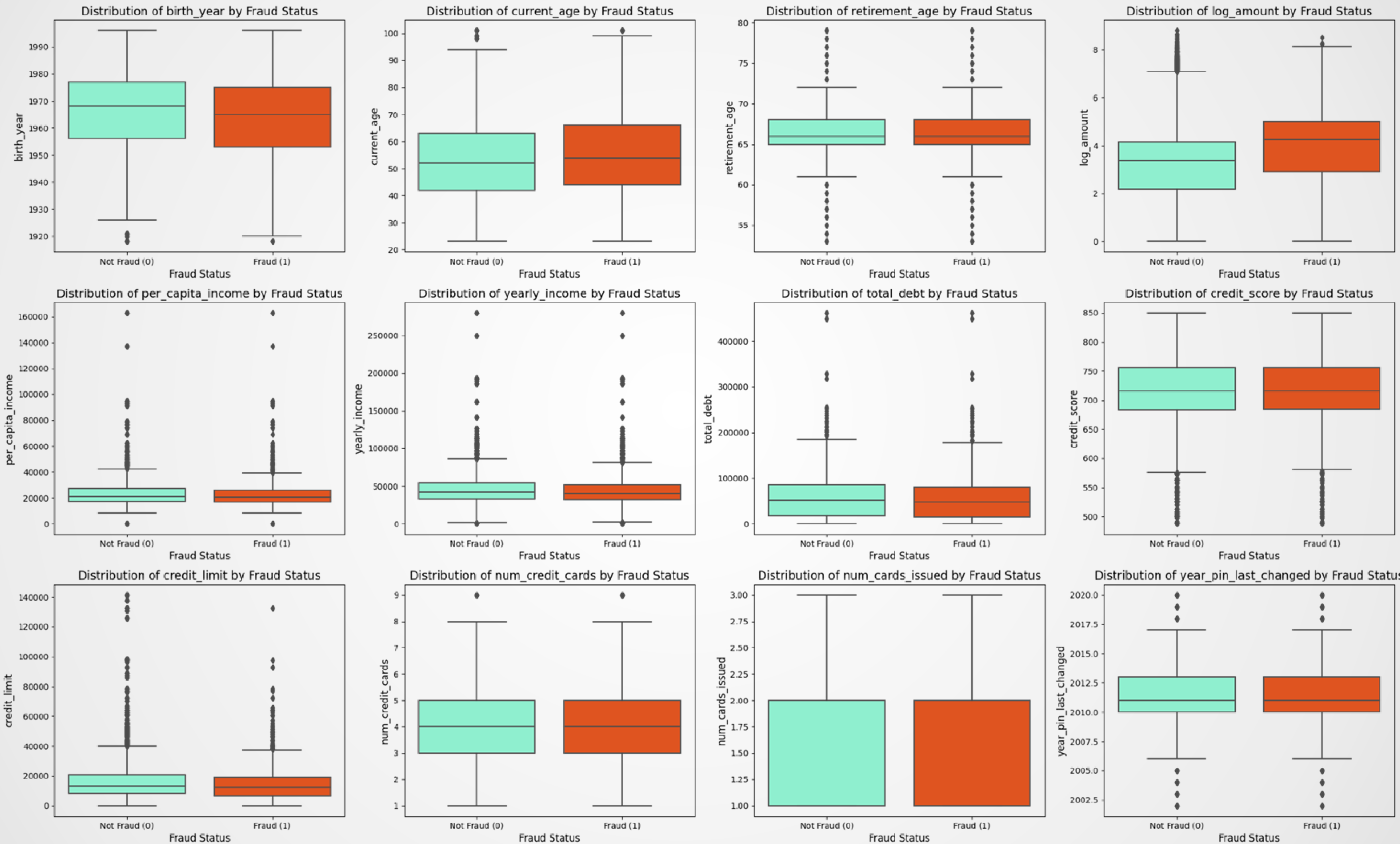
What's Special?

Potential Pattern of Fraudulent Data?

- High Amount
- Low Credit Limit



Correlation–Bivariate Box Plot



For benchmark model

- **Aim:** handle binary classification issue
- **Modeling the Logistic Regression**
  - numeric/ categorical variables
  - assumptions
  - estimation
- **Logistic Regression Assessment**
- **Compare the AUC and PR-AUC**

Confirm Variables
- Independent
- Dependent X_18 fraud_label

↓

Assumptions Checks
- Independence of observations
- Linearity of logit

↓

Handle Multicollinearity
- Drop one-hot encoding variables
- Check VIF and remove VIF>= 4

↓

Benchmark Model Assessment
- AUC
- PR-AUC

The assumption of logistic regression

- **Binary classification issue**
- **Few assumptions**
  - Independence of observations
  - Linearity of logit (Ignore)

- **Sample size (RULE of THUMB)**

  More than 400, 100 times of independents variables(29)
- **Training set:** 7,131,970
- **Testing set:** 1,782,994

Model Assessment– Interpretation & Performance

$$
\begin{aligned}
\operatorname{logit}(p) = &-7.911930 - 0.477340\,X_1 - 0.065822\,X_3 - 0.002348\,X_4 + 0.687795\,X_5 + 0.183640\,X_7 \\
&- 1.092849\,X_{10} - 0.080538\,X_{11} - 0.059312\,X_{15} - 1.214475\,X_{18} - 0.031945\,X_{20} + 0.022771\,X_{21} \\
&- 0.006862\,X_{26} - 0.338758\,X_{27} - 0.134344\,X_{29} + 0.000380\,X_{30} + 0.028046\,X_{31} + 0.190968\,X_{32} \\
&+ 0.004933\,X_{24} - 0.011546\,X_{35} + 0.016621\,X_{37} - 0.008741\,X_{39} - 0.269001\,X_{40} \\
&+ 0.056773\,X_{45} + 0.041741\,X_{49} - 0.085101\,X_{38}.
\end{aligned}
$$

Performance	Training set	Test set
ROC-AUC	0.8859	0.8862
PR-AUC	**0.0193**	**0.0257**

Model Building—workflow

Raw Data → **Exploratory Data Analysis + Data Preprocessing** → **Get X and y** → **Feature Generation** → **Feature Selection** → **ML Exection** → **Performance Evaluation**

Raw Data

Exploratory Data Analysis + Data Preprocessing
- Missing indicator
- Format conversion
- One-Hot Encoding
- Z Standardization
- Log Transformation
- Graph generation
- Principle Component Analysis

Get X and y
- X: 51 predictors
- y: fraud_label

Feature Generation
- RFM
- Domain Knowledge

Feature Selection
- Stepwise Selection
- Lasso
- Ridge
- Elastic Net

ML Exection
- LR
- XGB
- Light GBM

Performance Evaluation
- AUC
- PR-AUC

Feature Engineering

RFM framework

Feature Dimension	Feature Name	Definition
Recency	RecencyInterval	Time interval between consecutive transactions
Frequency	TxnFrequency_Xd	Number of transactions within a given X-day window (e.g. 7, 30, 60, 90 days)
Monetary	AmtDelta	Difference in the average of amounts

Domain knowledge

Feature Dimension	Feature Name	Definition
Location	Merchant_[Location]	Indicates the location of the transaction (e.g. online, us, eu, others)
Location	FirstTxnInRegion	Indicates whether this is the first transaction in a specific city, state, or region
Location	HighRiskMCC	Flags transactions associated with high-risk Merchant Category Codes (MCC)
Location	DifferentState	Indicate whether the transaction is made in a different state compared to the previous transaction for the same card
Others	TxnToLimitRatio	Ratio of transaction amount to the cardholder's credit limit

Variables Stepwise Selection guideline

1. **Model Fit Evaluation:**
   Use the −2 Log(Likelihood) value to assess overall model improvement.

2. **Stepwise Selection Procedure:**
   Iteratively add or remove variables based on statistical criteria.

3. **Entry and Removal Criteria:**
   Include variables with significant($p < 0.05$) LR test(add)/Wald statistic(remove) ; remove non-significant ones.

4. **Stopping Rule:**
   Continue until no excluded variable produces a statistically significant improvement in model likelihood.

5. **Goal:**
   Achieve a parsimonious model, good fit with minimal predictors.

Hair, J. F., Black, W. C., Babin, B. J., & Anderson, R. E. (2019).
Multivariate Data Analysis (8th ed.). Cengage Learning.

Classification Models Comparison

Model: Logistics Regression (LR), XGBoost (XGB) and Light GBM (LGBM)

Index	Model	Features	AUC Train	AUC Test	PR AUC Train	PR AUC Test
0	LR	Xall	0.8859	0.8862	0.0193	0.0257
1	LR	Forward + backward stepwise	0.884	0.8374	0.0224	0.0256
2	LR	Lasso	0.8841	0.8378	0.0223	0.0256
3	LR	Ridge	0.8841	0.8378	0.0223	0.0256
4	LR	Elastic net	0.884	0.8378	0.0223	0.0256
5	LR	Xraw + Xrfm + Xdk + elastic net	0.9408	0.952	0.1068	0.0506
6	LR	Xraw + Xrfm + Xdk– identifiers	0.9143	0.9025	0.0811	0.0695
7	XGB	Xall	0.9900	0.9800	0.8200	0.4200
8	XGB	Xraw + Xrfm + Xdk	0.9800	0.9900	0.3900	0.3400
9	LGBM	Xall	0.9900	0.9600	0.8000	0.3900
10	LGBM	Xraw + Xrfm + Xdk	0.9300	0.9600	0.3200	0.2900

Conclusion

- **RFM behavioral signals:** Recency, Frequency, Monetary intensity
- **Anomalies pattern:** sudden activity spikes, high transaction frequency, abnormal amounts
- **Contextual risk cues:** limit-ratio features, high-risk MCCs, cross-region indicators
- **Interpretability:** transparent feature meanings, traceable risk patterns
- **Operational insights:** spending surges, regional deviations
- **Scalable design:** continuous monitoring, adaptable to behavior shifts

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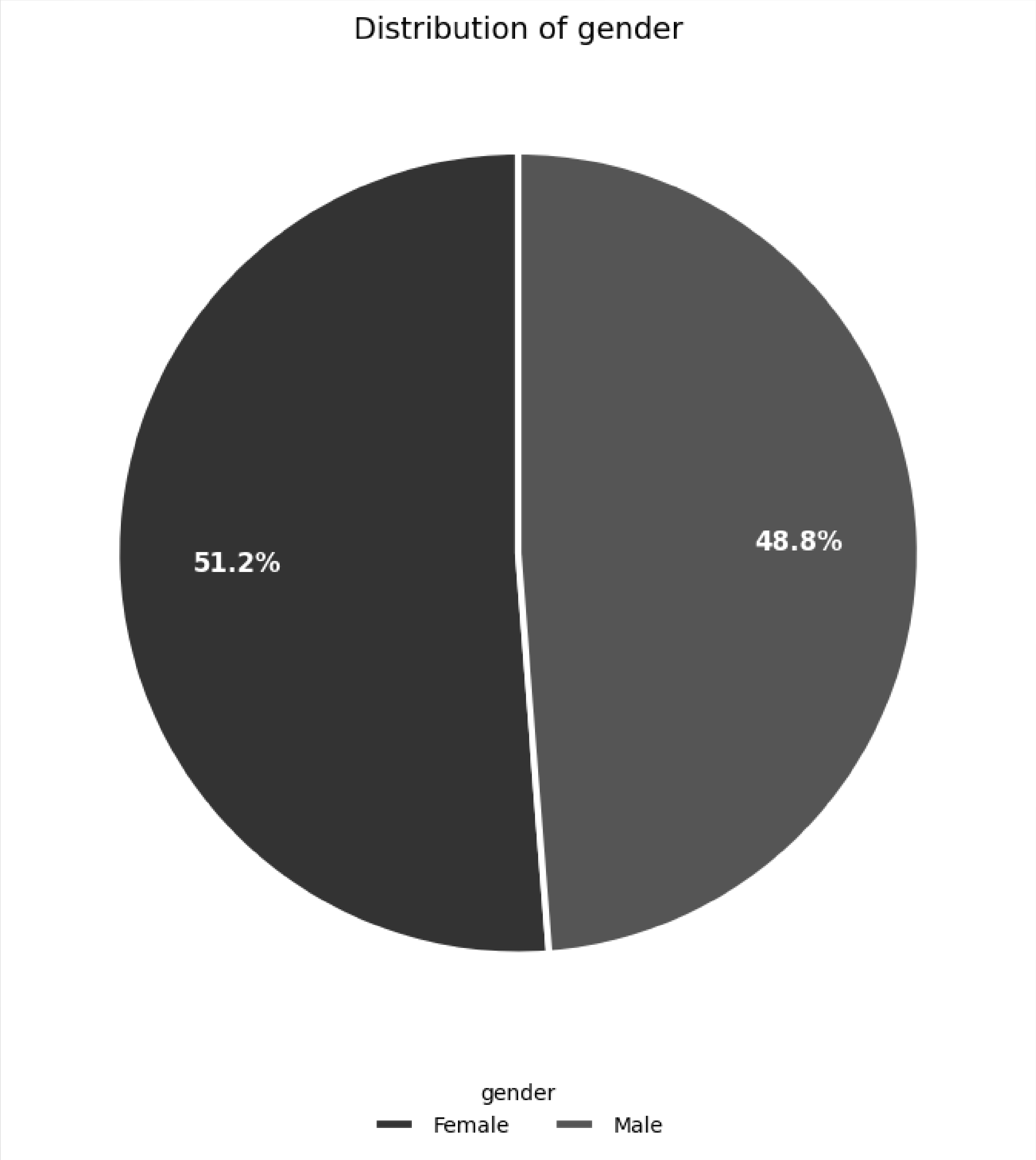
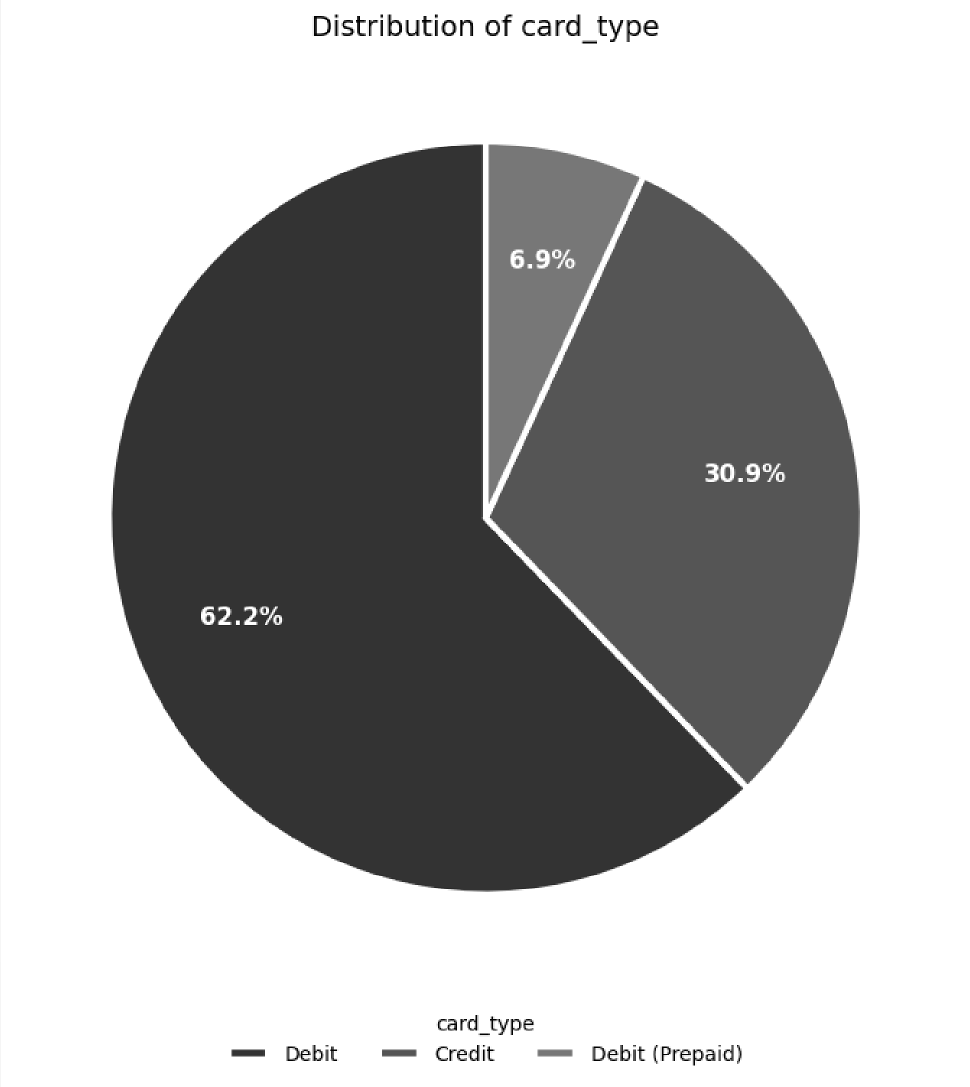
Appendix

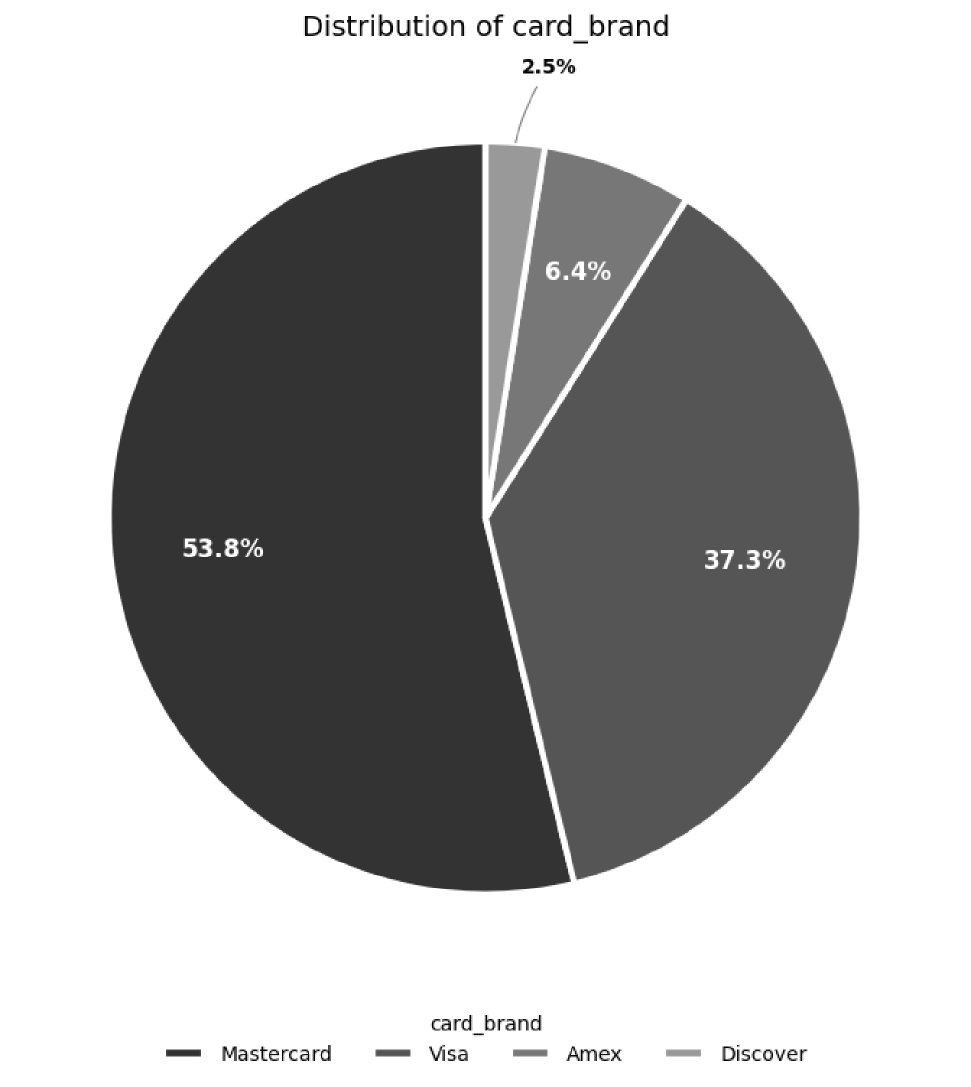
Categorical Variables — Consumers' Gender



Gender	Frequency
Female	6,815,916 (51.22%)
Male	6,489,999 (48.78%)

Categorical Variables—Card Information

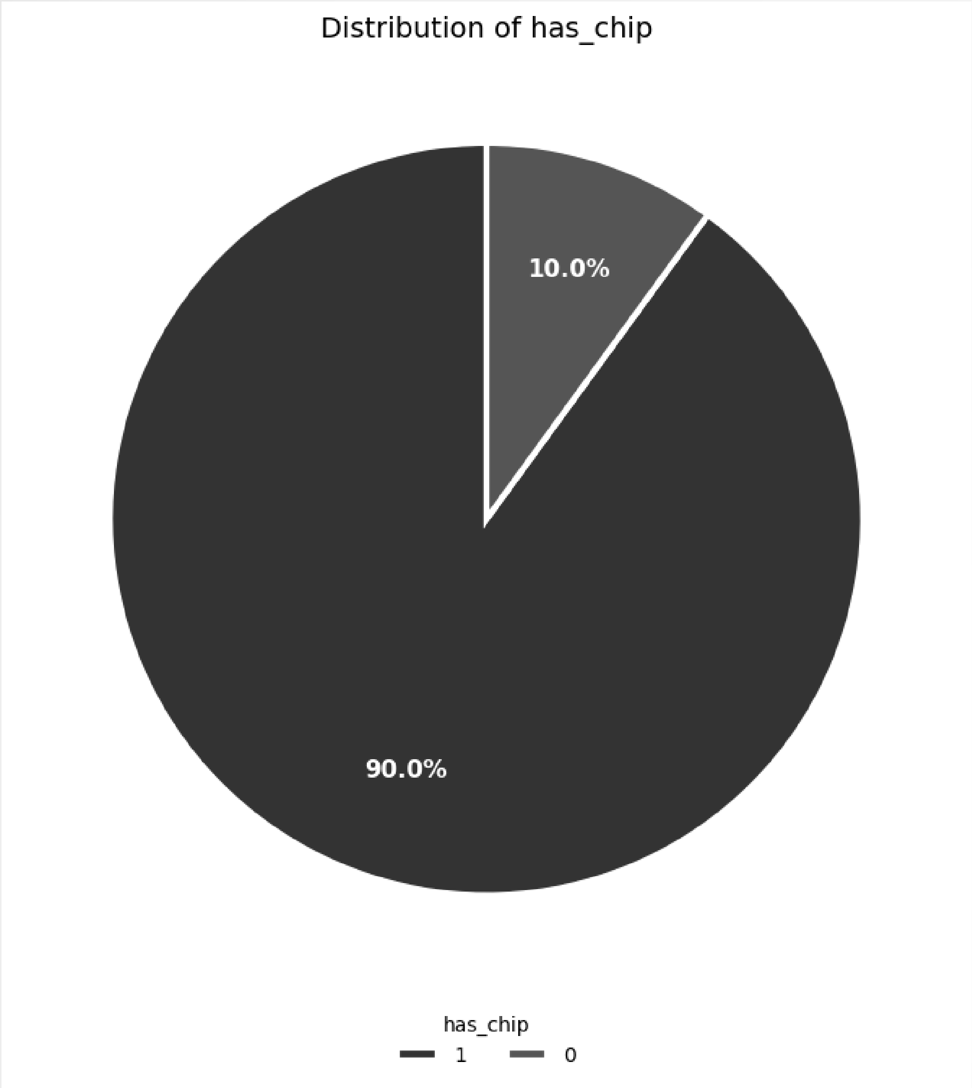


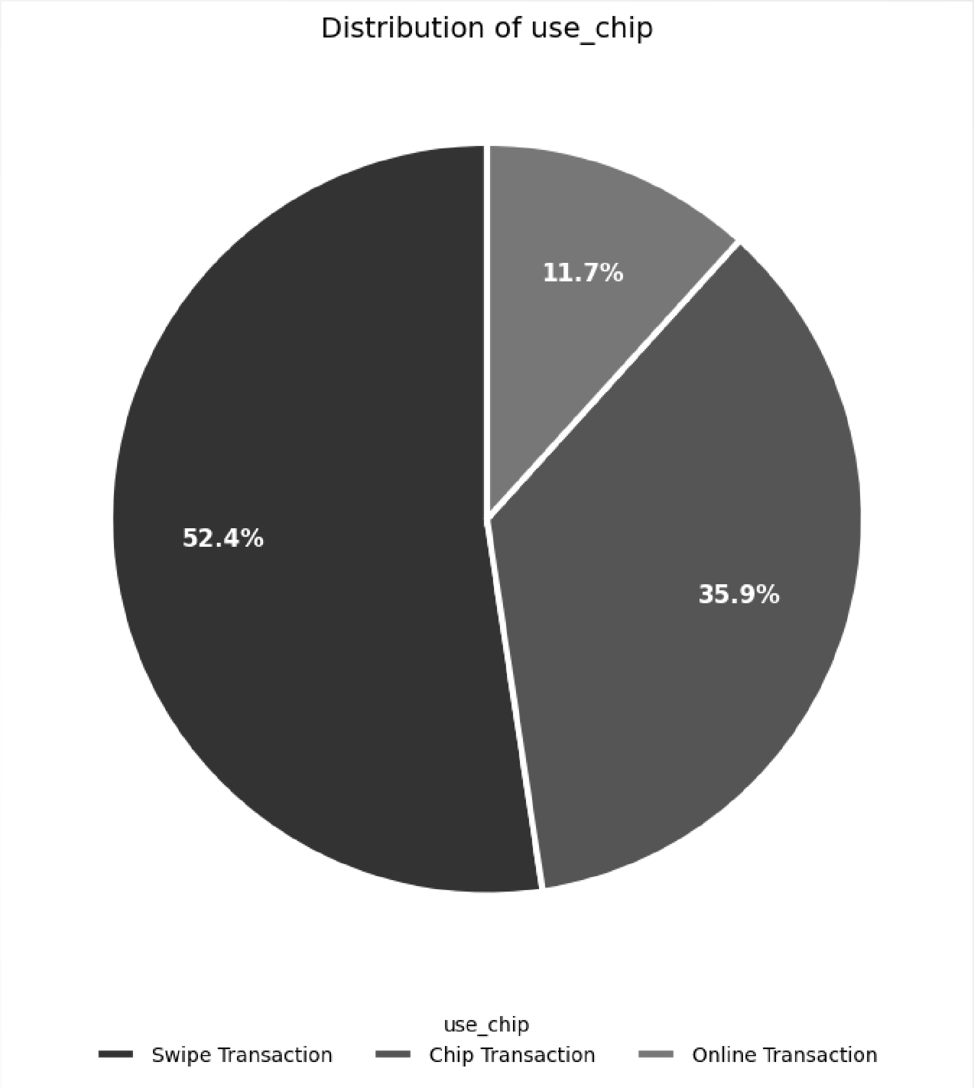


Card Type	Frequency
Debit	8,280,996 (62.24%)
Credit	4,109,189 (30.88%)
Debit (Prepaid)	915,730 (6.88%)

Card Brand	Frequency
Mastercard	7,157,399 (53.79%)
Visa	4,957,563 (37.26%)
Amex	854,490 (6.42%)
Discover	336,463 (2.53%)

Categorical Variables—Chip

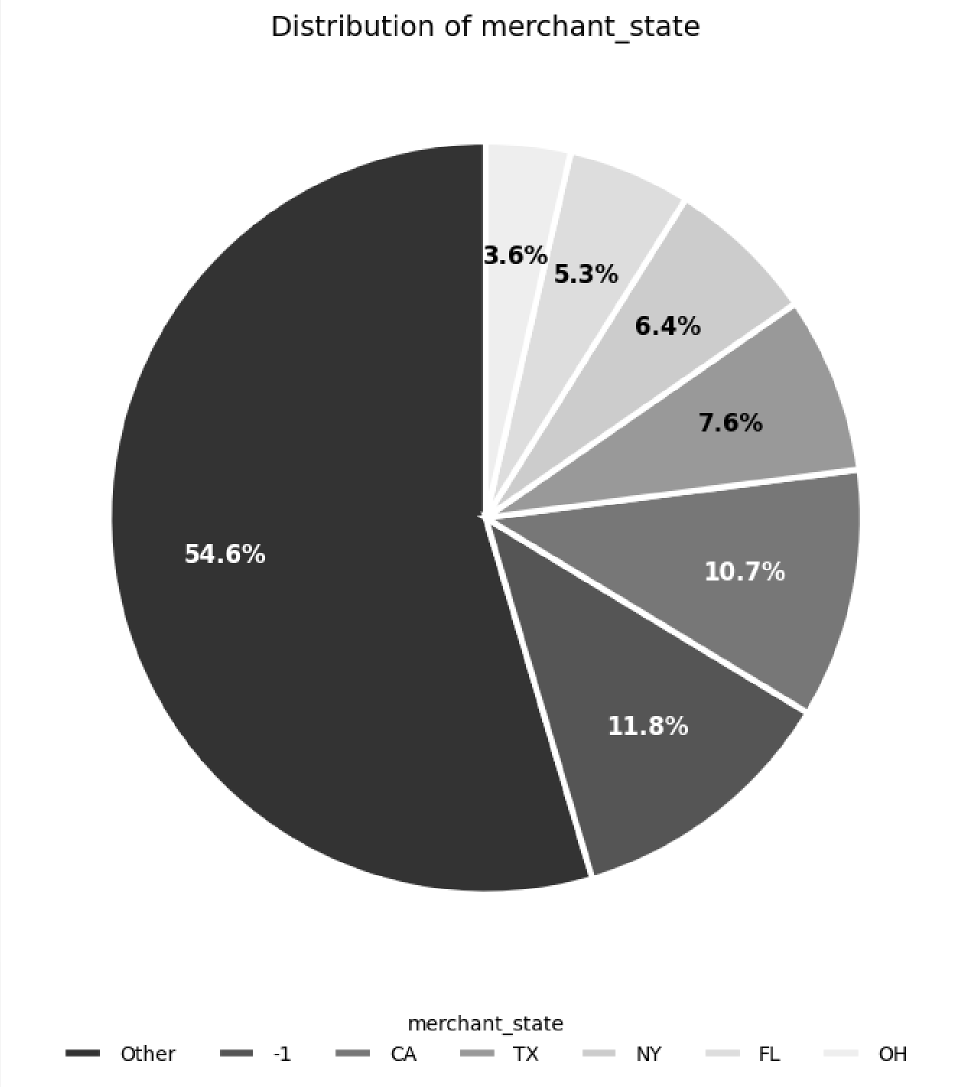


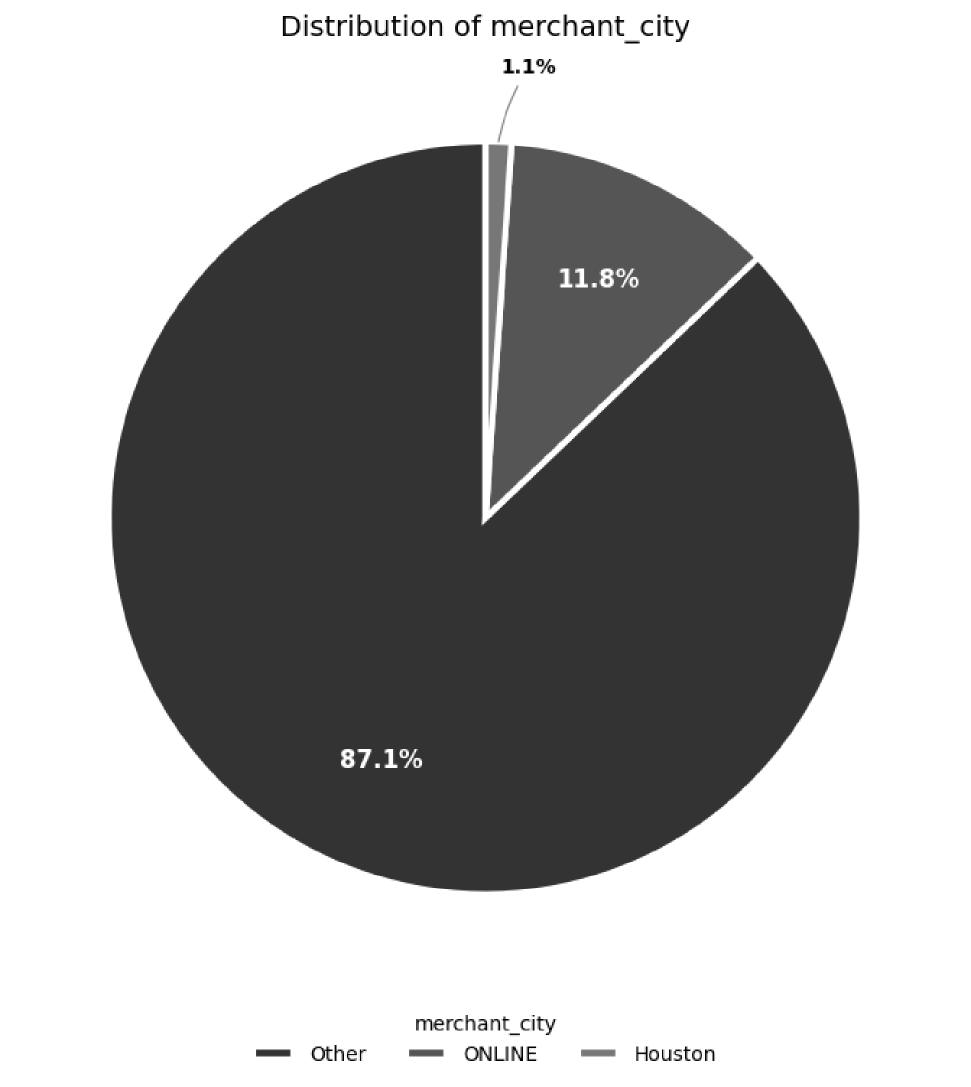


Has Chip	Frequency
Yes (1)	11,974,795 (90.00%)
No (0)	1,331,120 (10.00%)

Use Chip	Frequency
Swipe Transaction	6,967,185 (52.36%)
Chip Transaction	4,780,818 (35.93%)
Online Transaction	1,557,912 (11.71%)

Categorical Variables—Merchant Location/Online

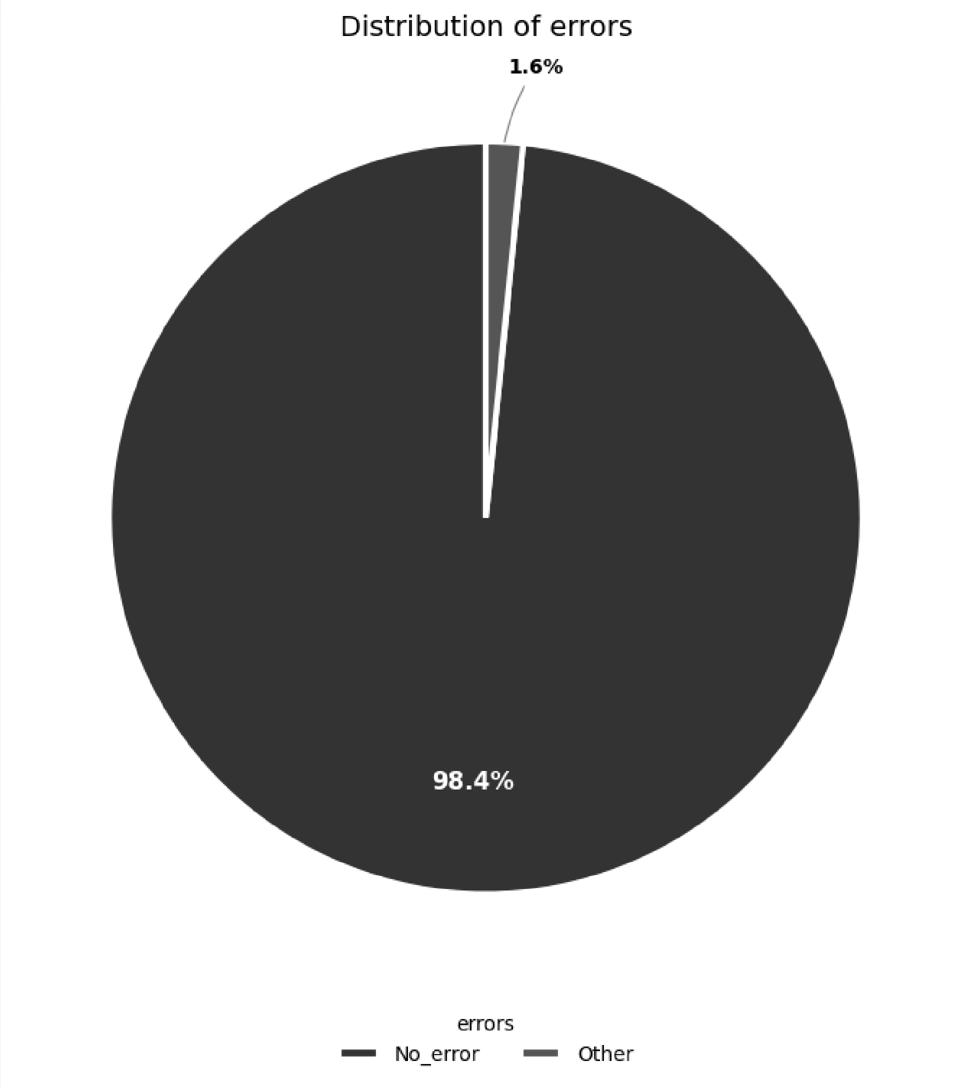


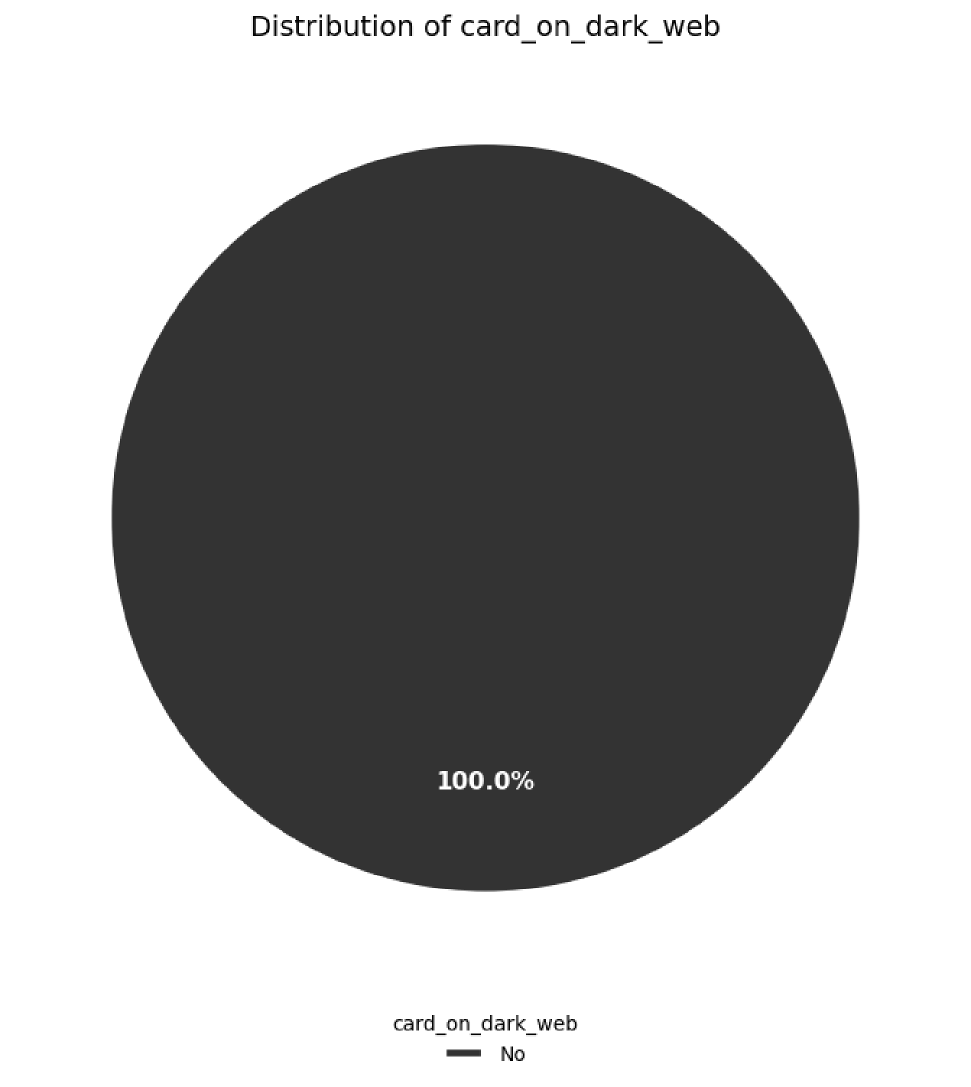


Merchant State	Frequency
Online	1,563,700 (11.75%)
CA	1,427,087 (10.73%)
TX	1,010,207 (7.59%)
NY	857,510 (6.44%)

Merchant City	Frequency
Online	1,563,700 (11.75%)
Houston	146,917 (1.10%)
Miami	87,388 (0.66%)
Brooklyn	84,020 (0.63%)

Categorical Variables—Abnormal Transaction

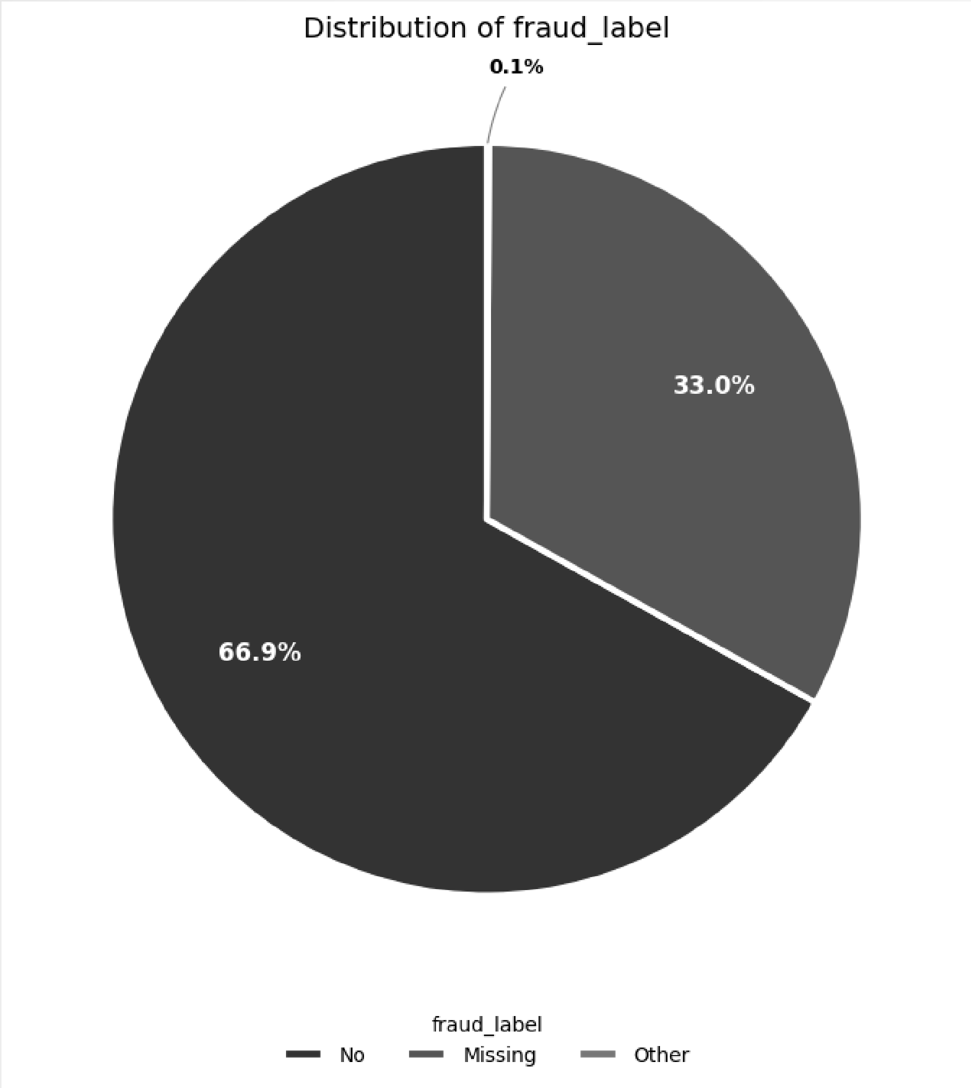


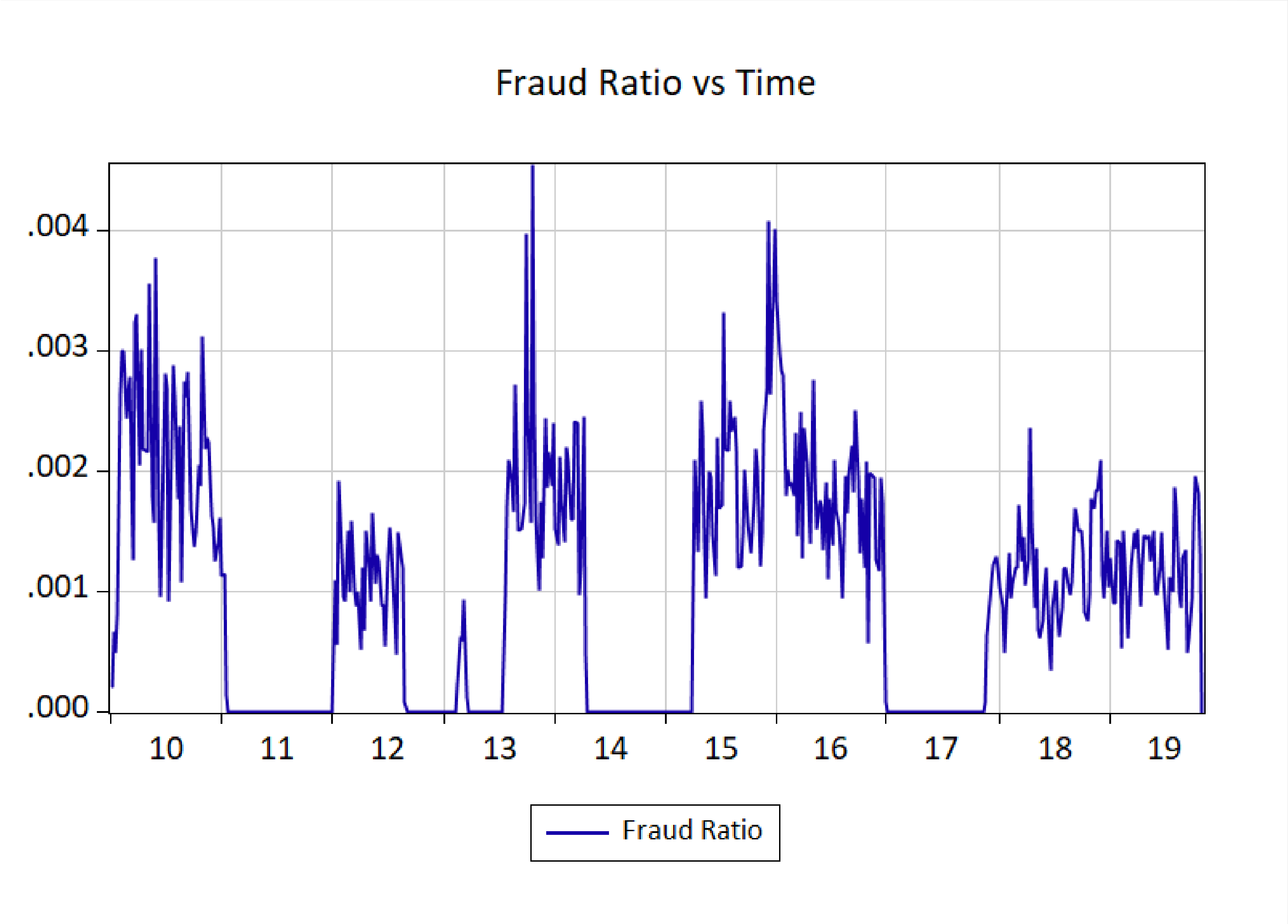


Errors	Frequency
No Error	13,094,522 (98.41%)
Insufficient Balance	130,902 (0.98%)
Bad Pin	32,119 (0.24%)
Technical Glitch	26,271 (0.20%)

Card on Dark Web	Frequency
No (0)	13,305,915 (100%)
Yes (1)	0 (0%)

Categorical Variables—Fraud Label

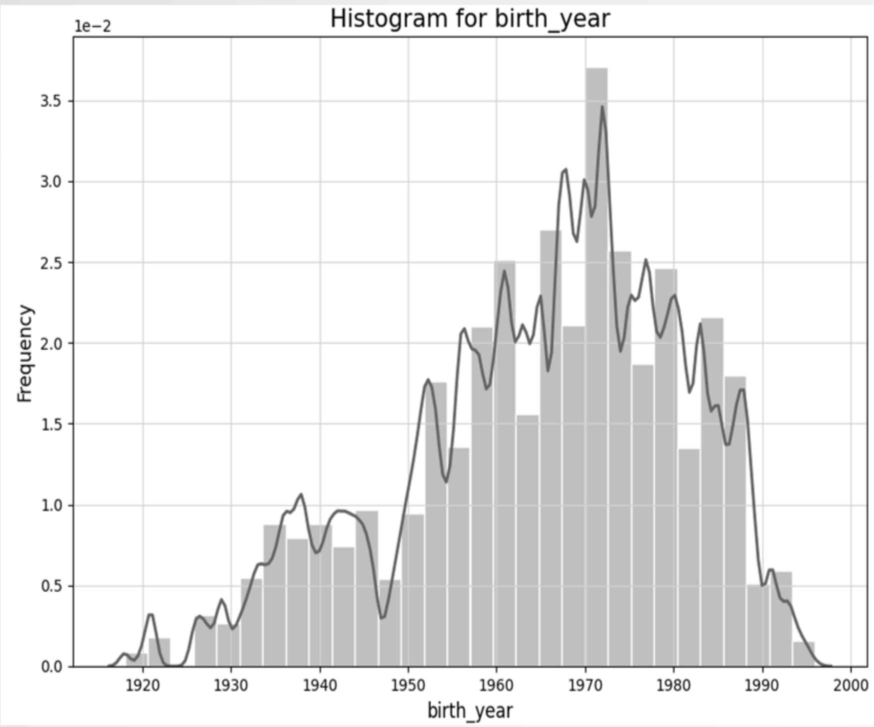


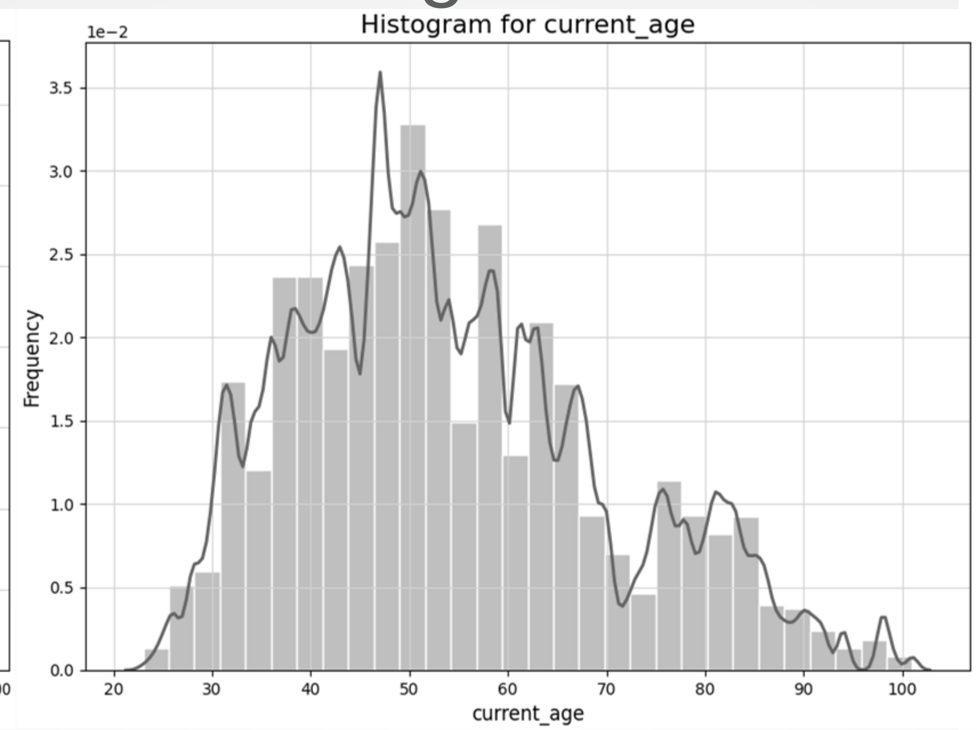


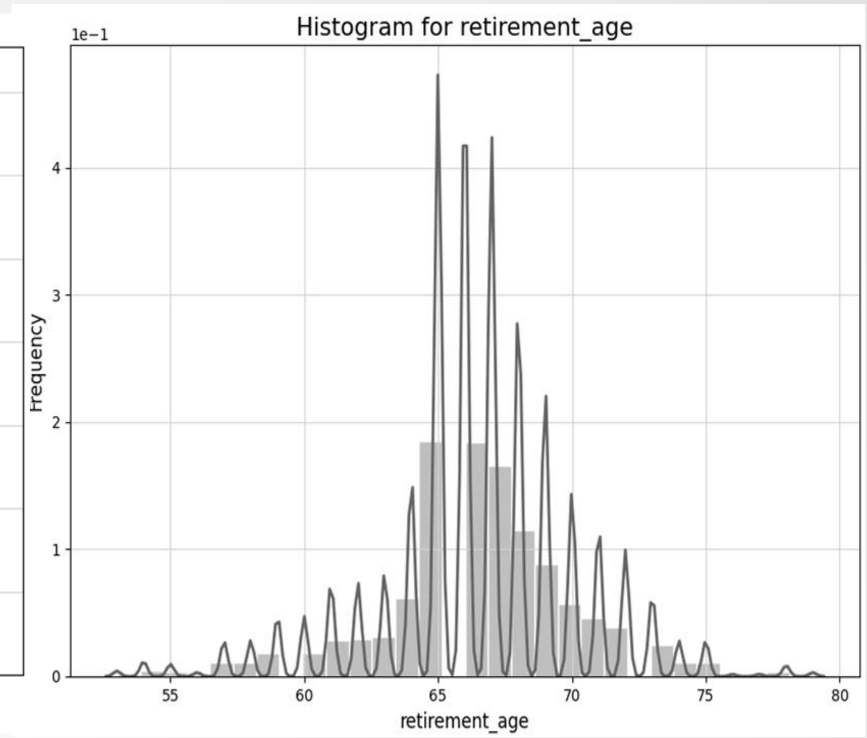
Fraud Label	Frequency
No	8,901,631 (66.90%)
Missing	4,390,952 (33.00%)
yes	13,332 (0.10%)

Fraud w/o Missing	Frequency
No	8,901,631 (99.85%)
Yes	13,332 (0.15%)

Continuous Variables—Client Age

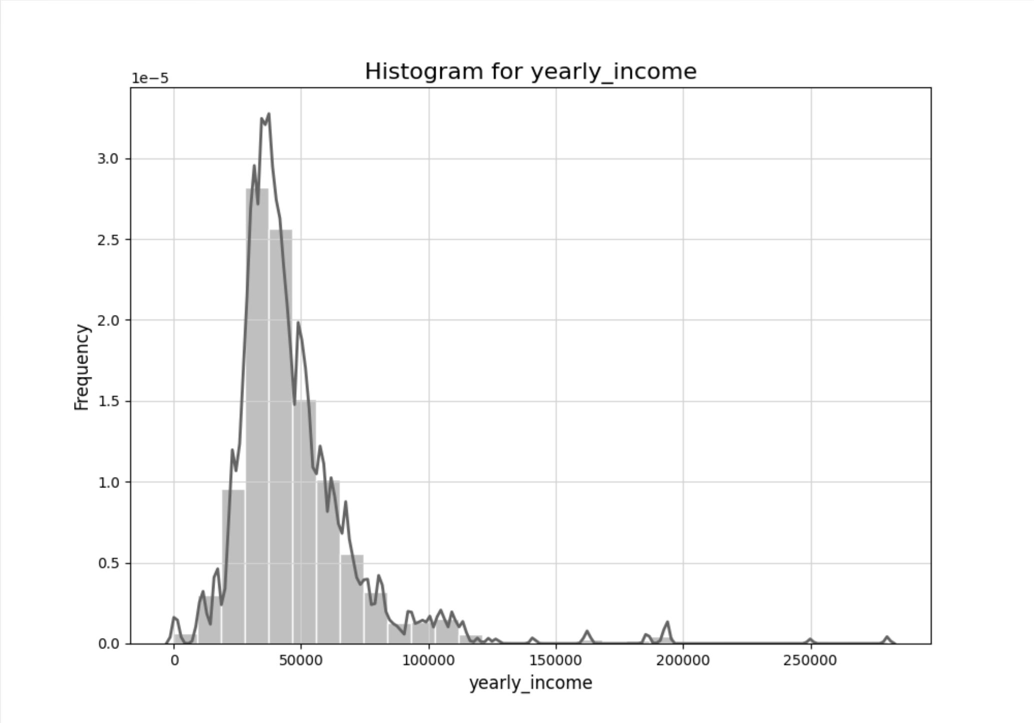


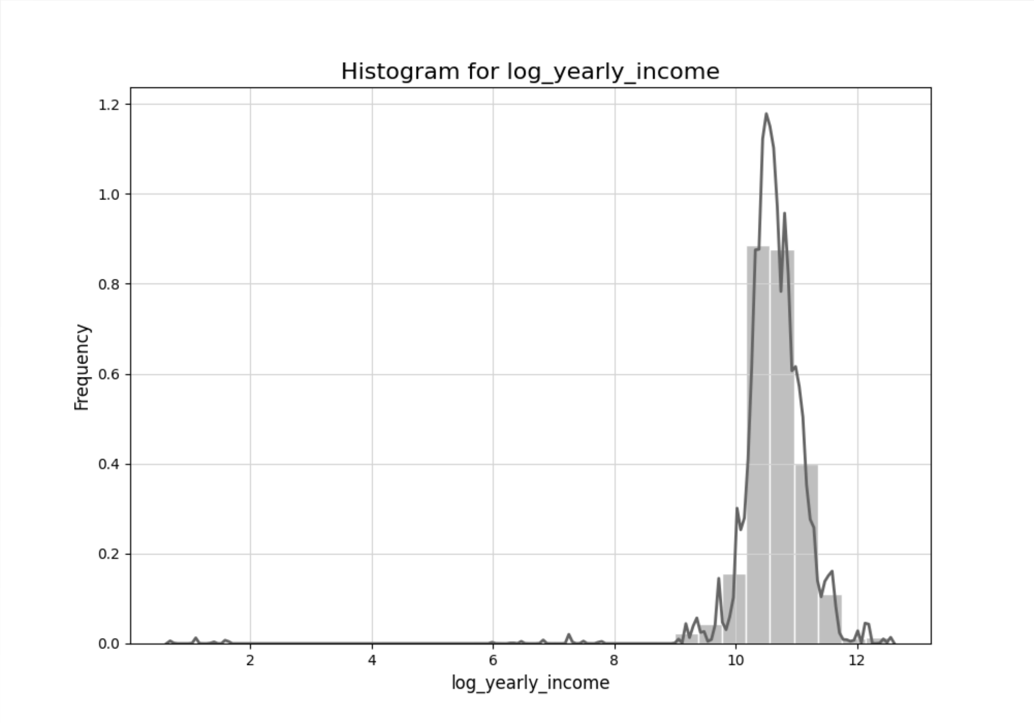




Statistics	birth_year	current_age	retirement_age
count	13,305,915	13,305,915	13,305,915
mean	1965.16	54.03	66.49
std	15.72	15.73	3.59
min	1918	23	53
25%	1956	42	65
50%	1968	52	66
75%	1977	63	68
max	1996	101	79
skewness	-0.55	0.55	-0.32
kurtosis	-0.27	-0.28	1.38

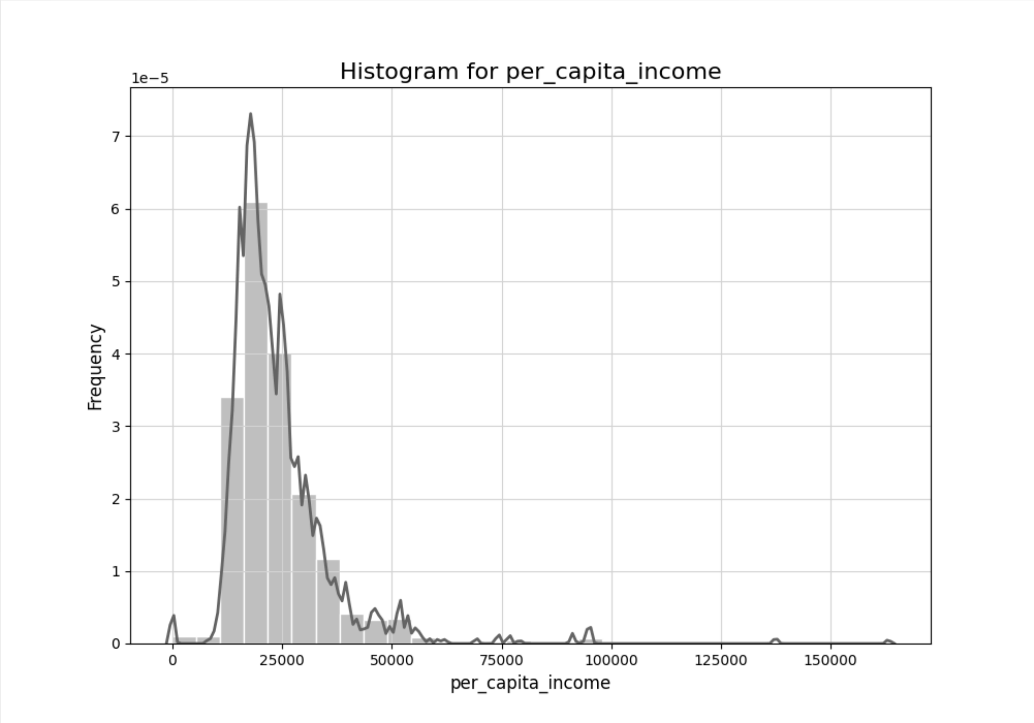
Continuous Variables—Income

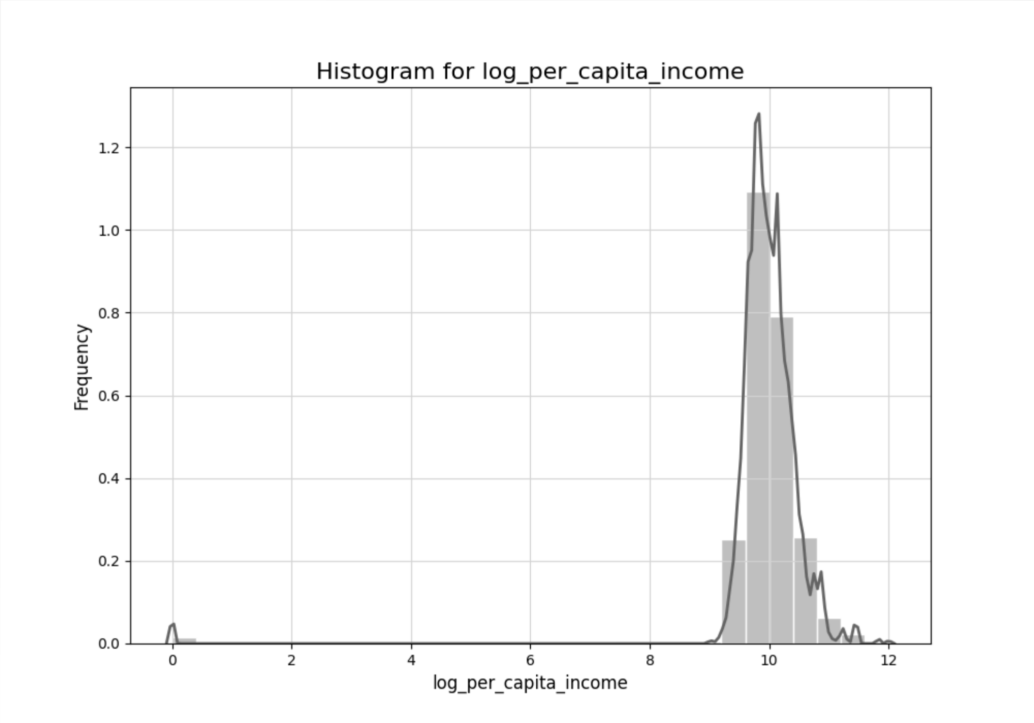




Statistics	yearly_income	log_yearly_income
count	13,305,915	13,305,915
mean	46,683.69	10.63
std	24,445.09	0.65
min	1	0.69
25%	32,817	10.40
50%	41,069	10.62
75%	54,013	10.90
max	280,199	12.54
skewness	3.11	-7.16
kurtosis	18.07	98.51

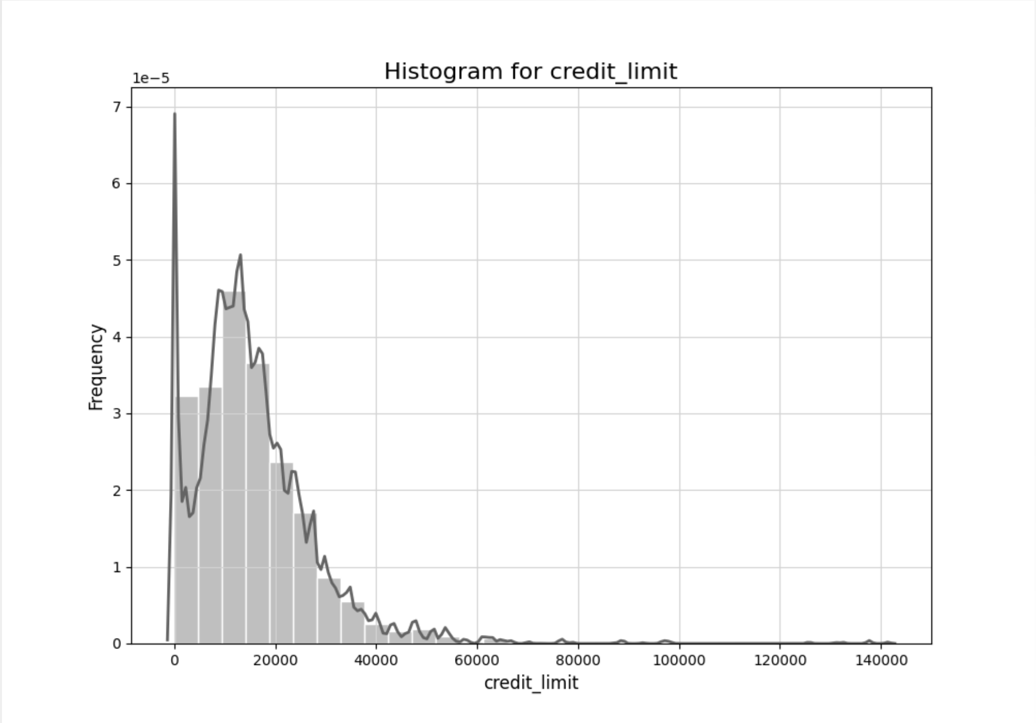
Continuous Variables—Income

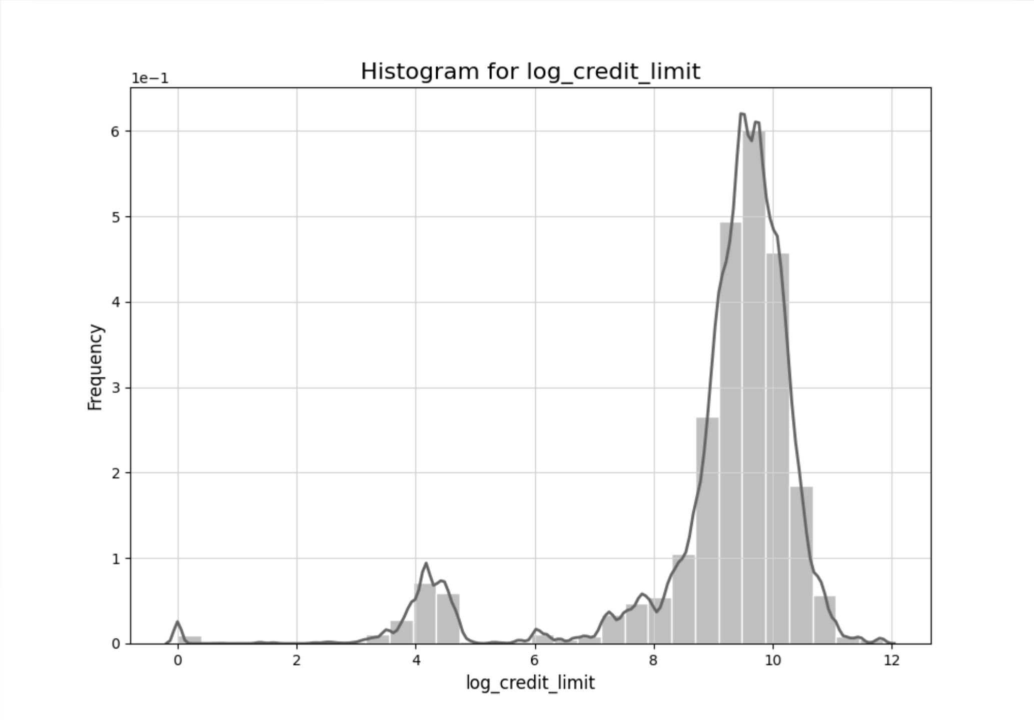




Statistics	per_capita_income	log_per_capita_income
count	13,305,915	13,305,915
mean	23,979.65	9.96
std	11,967.12	0.83
min	0	0
25%	17,113	9.75
50%	21,156	9.96
75%	27,308	10.21
max	163,145	12.00
skewness	3.72	-9.38
kurtosis	26.21	110.55

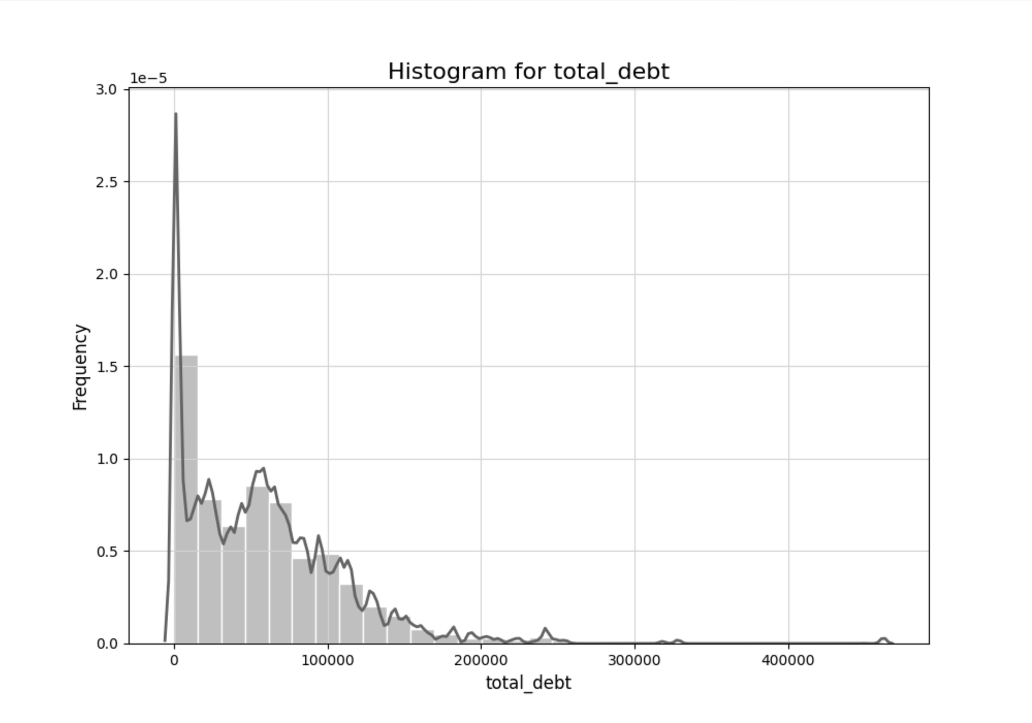
Continuous Variables—Credit Limit

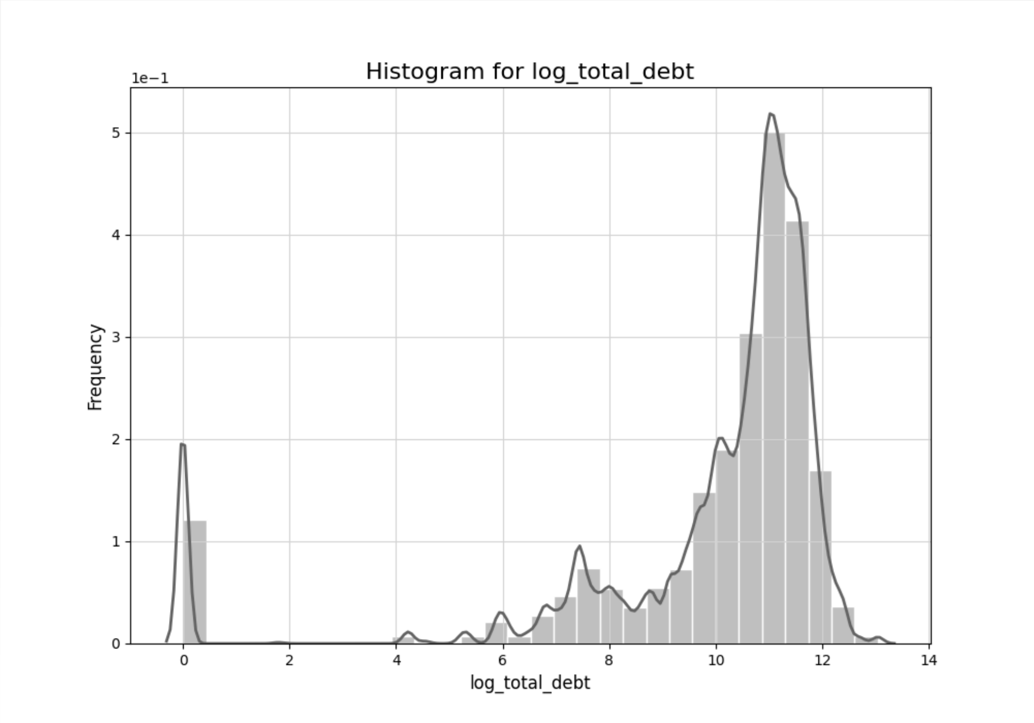




Statistics	credit_limit	log_credit_limit
count	13,305,915	13,305,915
mean	15,549.83	9.08
std	12,181.06	1.66
min	0	0
25%	8,100	9.00
50%	13,455	9.51
75%	20,839	9.94
max	141,391	11.86
skewness	2.42	-2.49
kurtosis	14.52	6.61

Continuous Variables—Debt

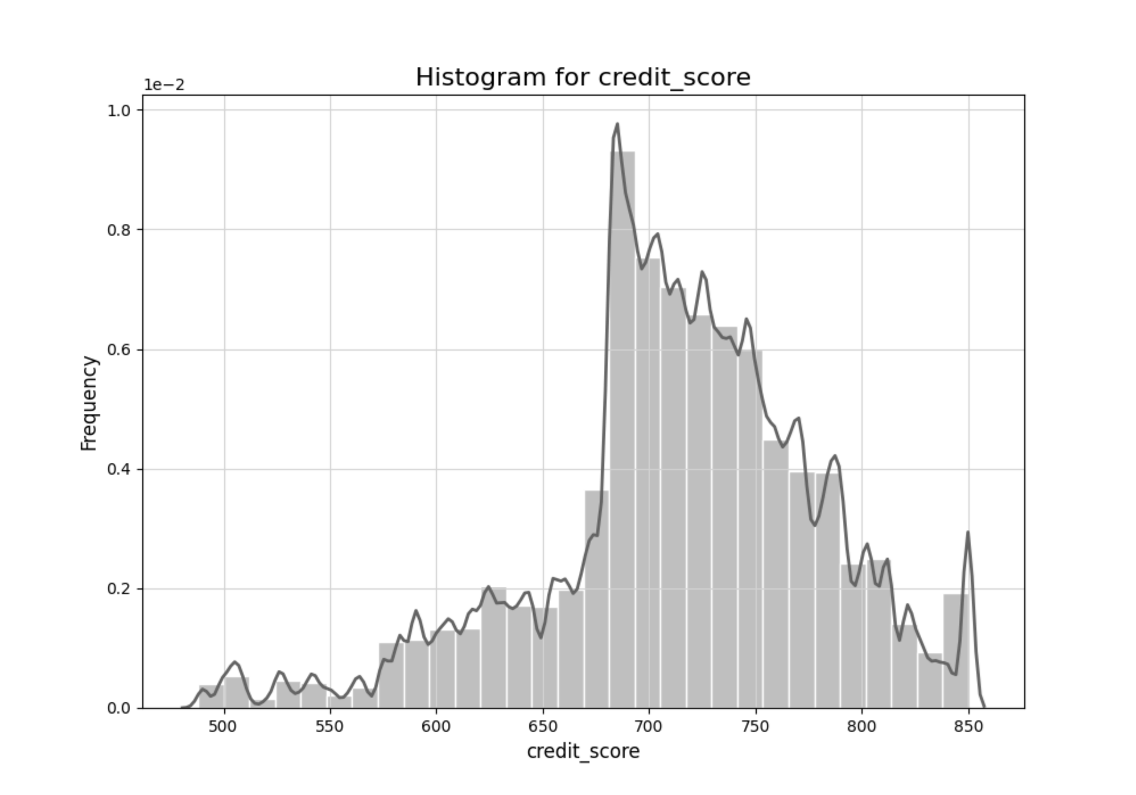




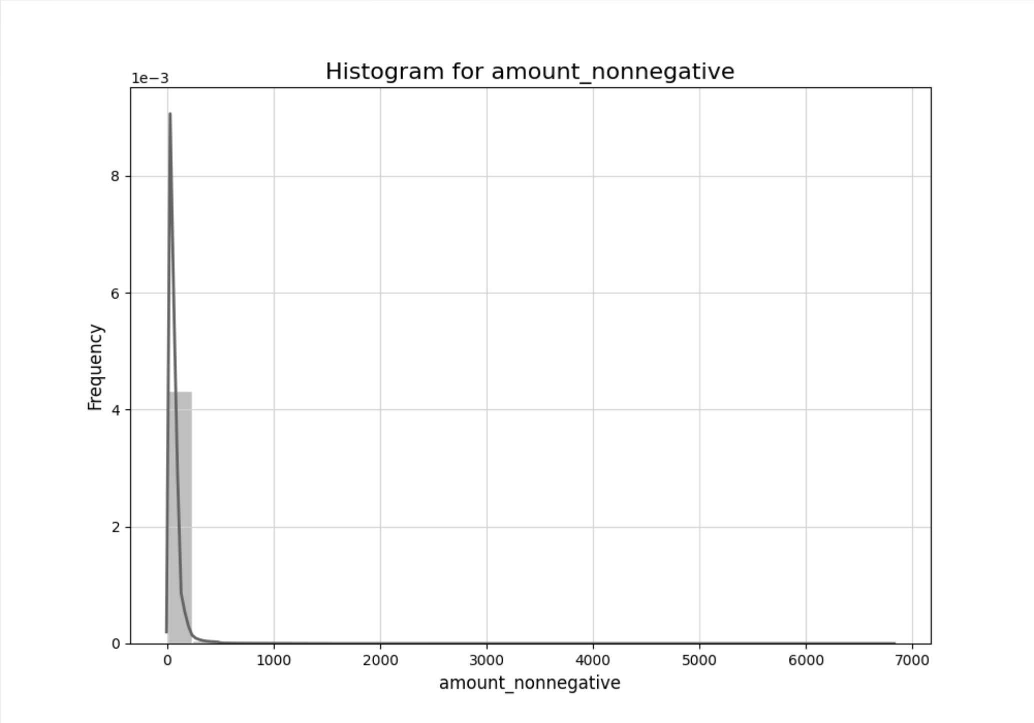
Statistics	total_debt	log_total_debt
count	13,305,915	13,305,915
mean	58,022.07	9.88
std	52,100.96	2.73
min	0	0
25%	16,437	9.71
50%	51,679	10.85
75%	85,160	11.35
max	461,854	13.04
skewness	1.74	-2.56
kurtosis	7.05	6.39

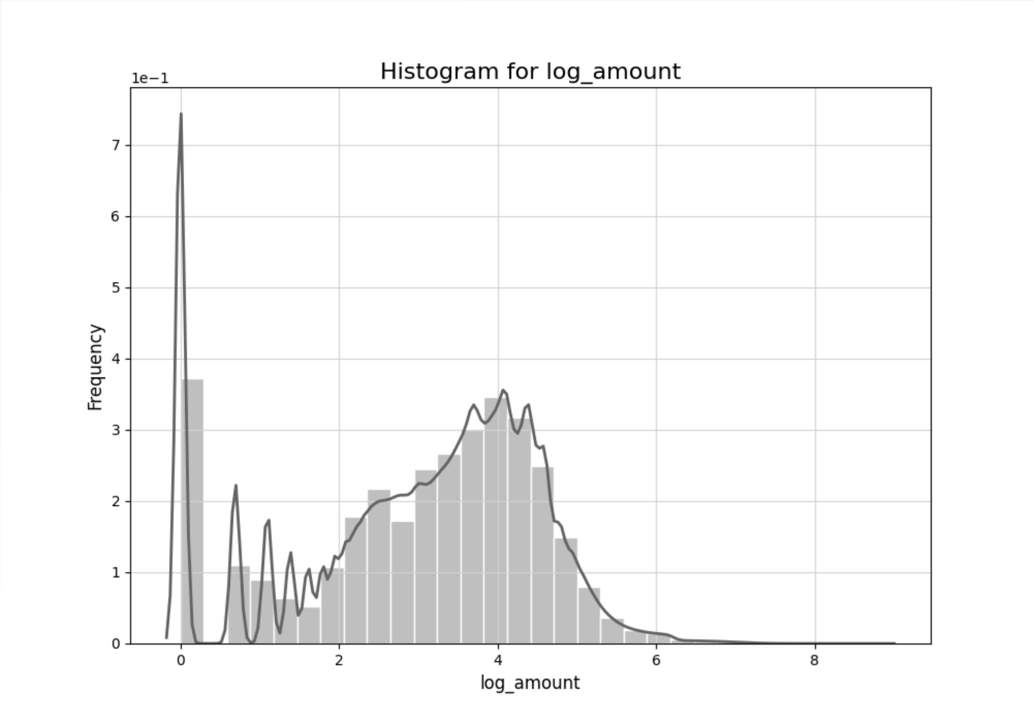
Continuous Variables—Credit Score

Statistics	credit_score
count	13,305,915
mean	713.91
std	65.80
min	488
25%	684
50%	716
75%	756
max	850
skewness	-0.60
kurtosis	0.82

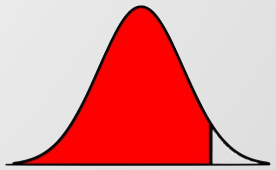


Continuous Variables—Amount

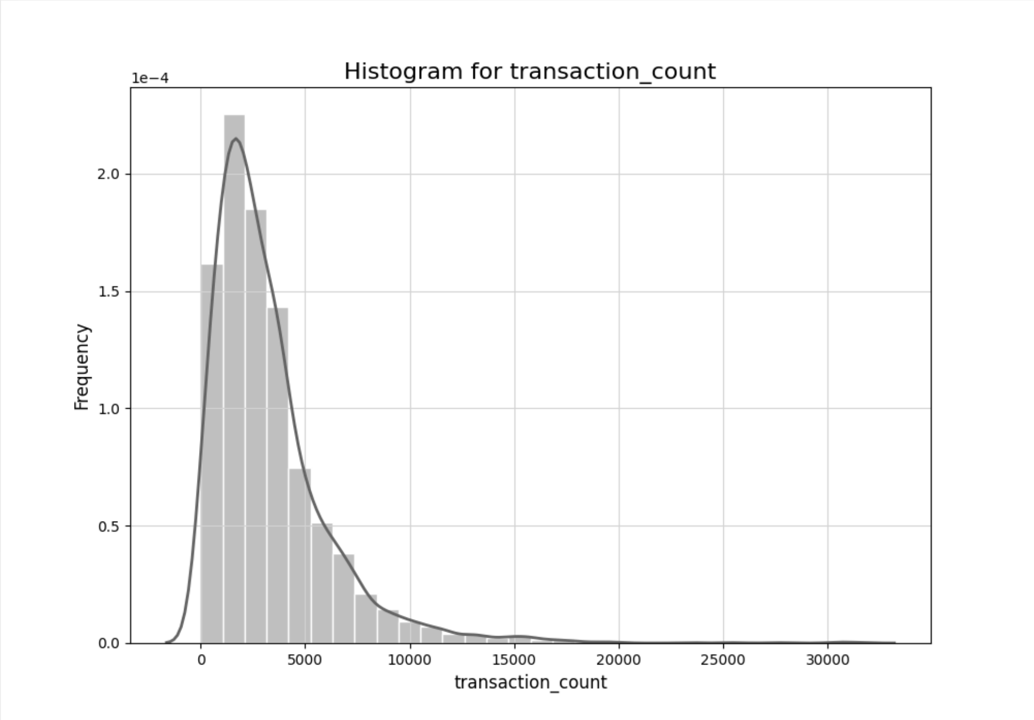


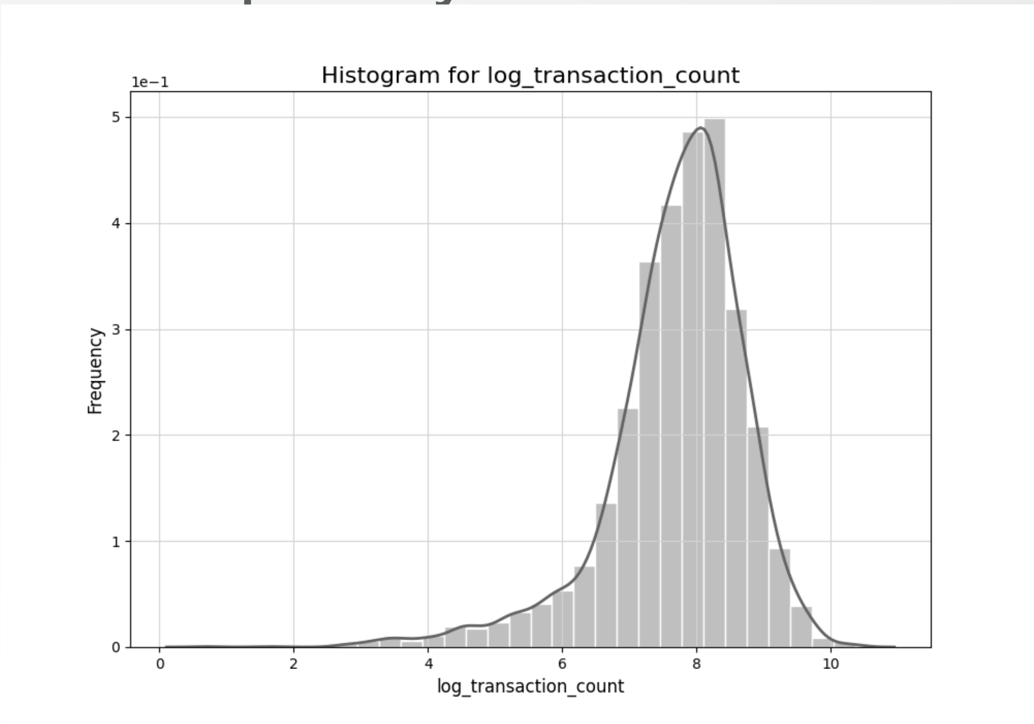


Statistics	amount/amount_nonnegative	log_amount*
count	13,305,915	13,305,915
mean	42.55/47.70	3.00
std	81.64/72.97	1.54
min	-500/1	0
25%	8	2.08
50%	28	3.33
75%	63	4.14
max	6,820	8.83
skewness	5.27/8.13	-0.58
kurtosis	111.28/168.48	-0.49



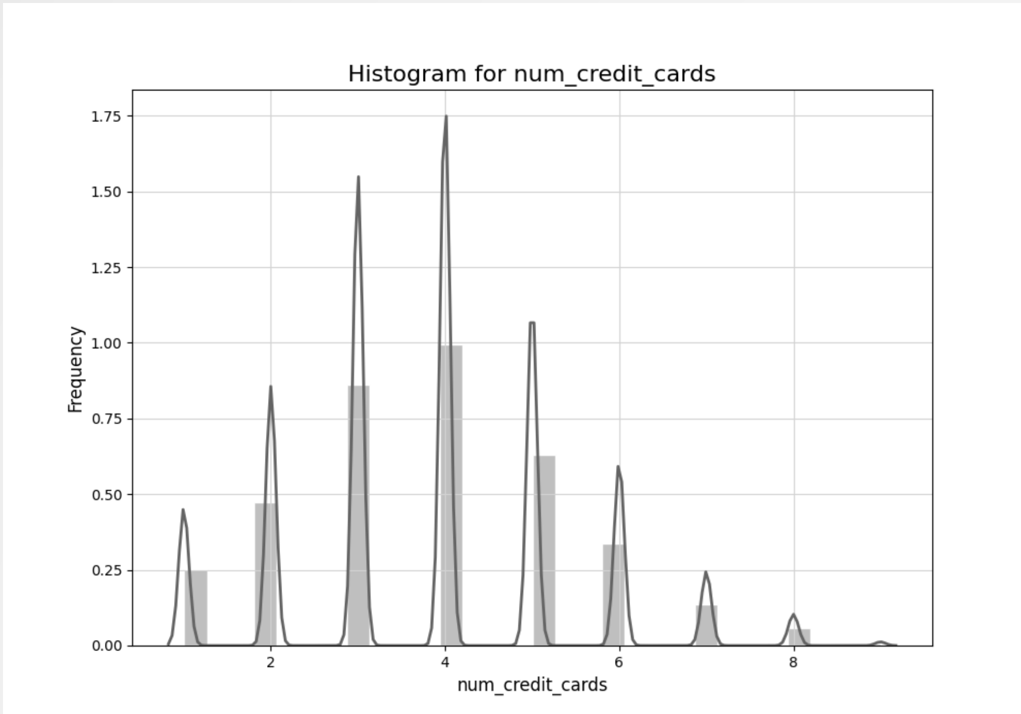
Continuous Variables—Transaction Frequency*

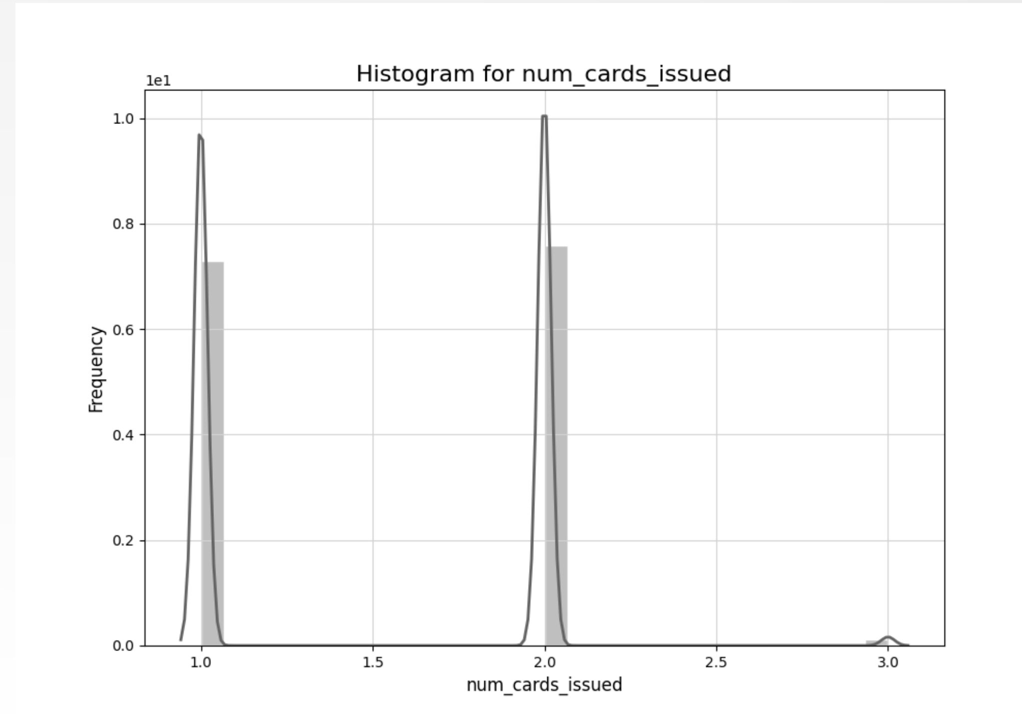




Statistics	transaction_count	log_transaction_count
count	4,071	4,071
mean	3,268.46	7.70
std	2,866.22	1.03
min	1	0.69
25%	1,419	7.26
50%	2,558	7.85
75%	4,166.5	8.34
max	31,552	10.36
skewness	2.58	-1.34
kurtosis	12.59	3.47

Continuous Variables—Number of Cards

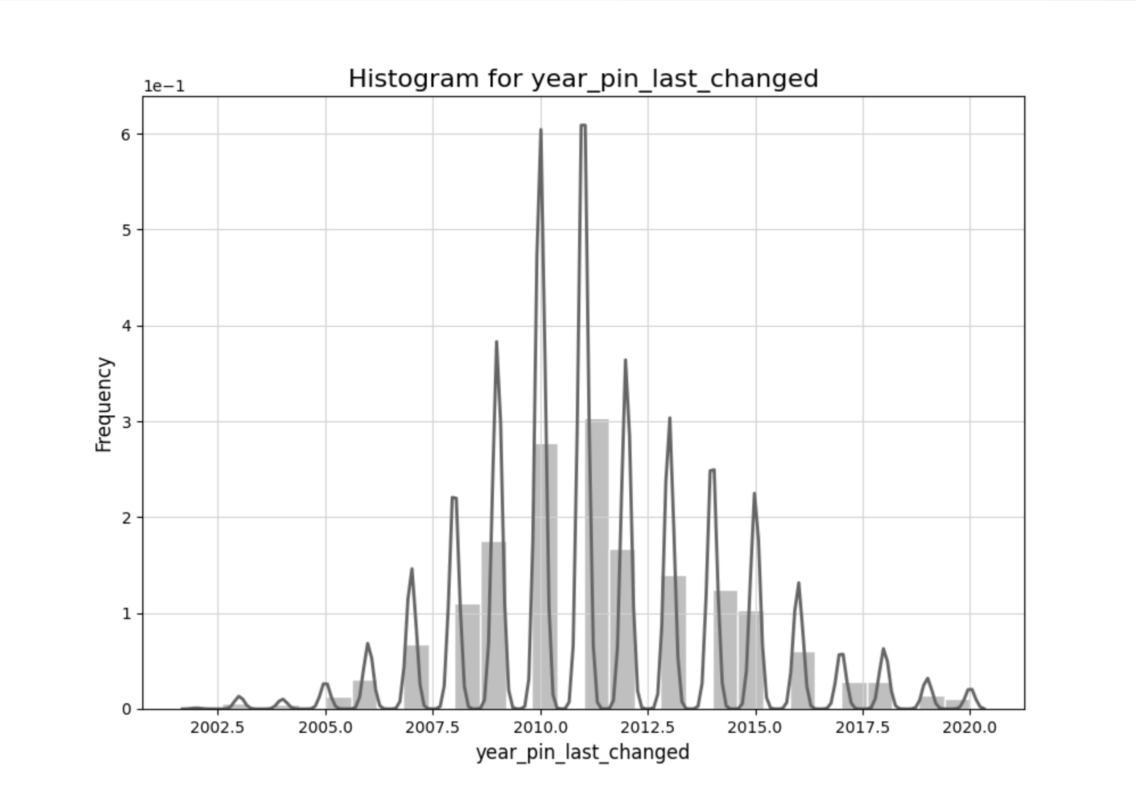




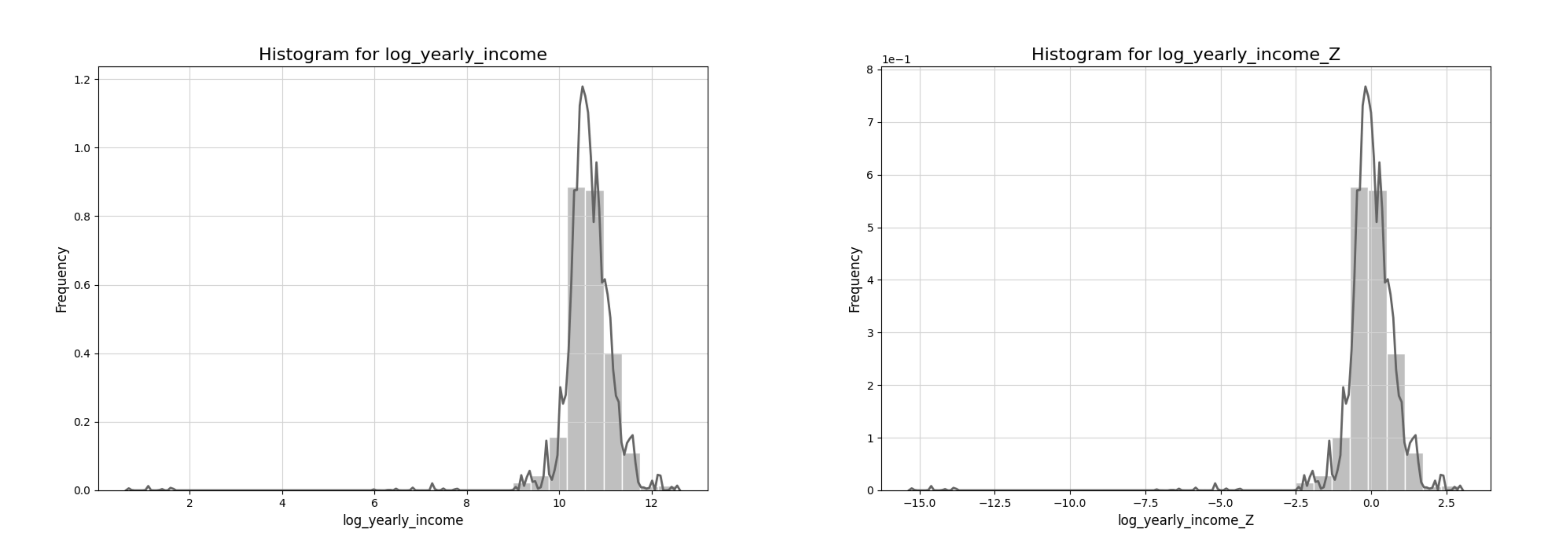
Statistics	num_credit_cards*	num_card_issued
count	13,305,915	13,305,915
mean	3.84	1.52
std	1.57	0.52
min	1	1
25%	3	1
50%	4	2
75%	5	2
max	9	3
skewness	0.31	0.08
kurtosis	-0.08	-1.57

Continuous Variables—Year Pin

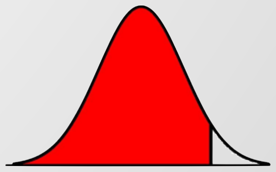
Statistics	year_pin_last_changed
count	13,305,915
mean	2,011.34
std	2.89
min	2002
25%	2010
50%	2011
75%	2013
max	2020
skewness	0.36
kurtosis	0.29



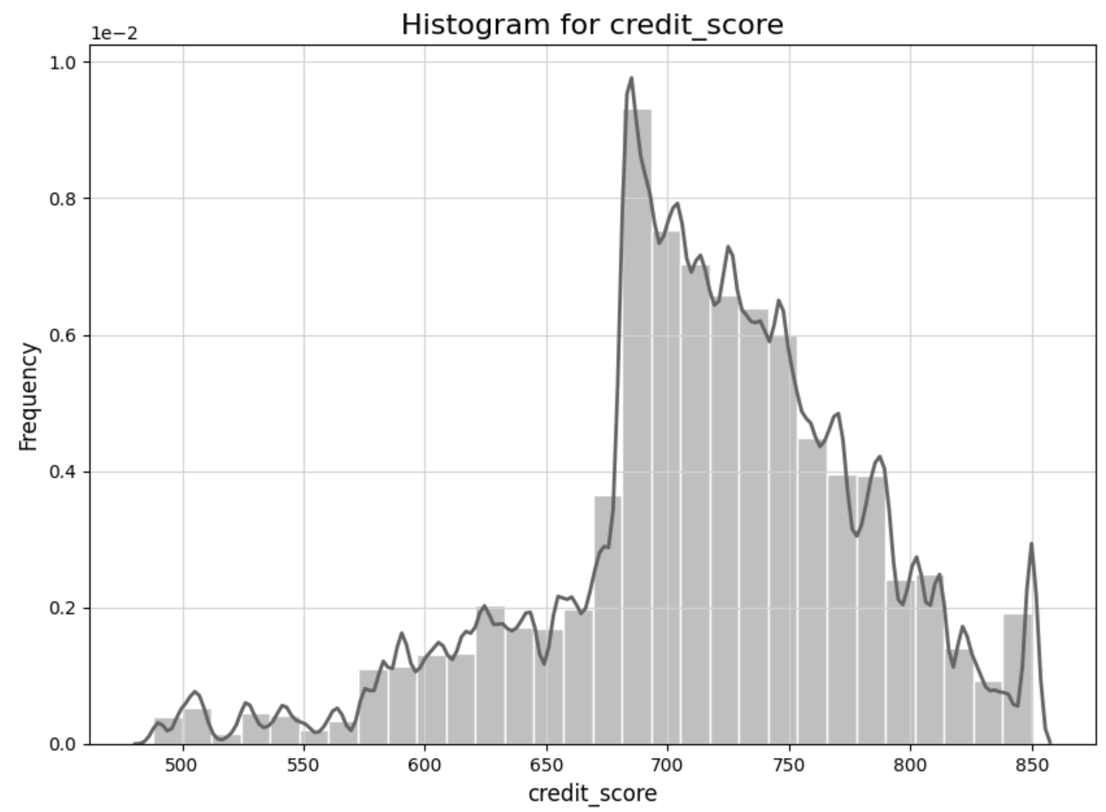
Scaling–Z score for Logarithm Variable

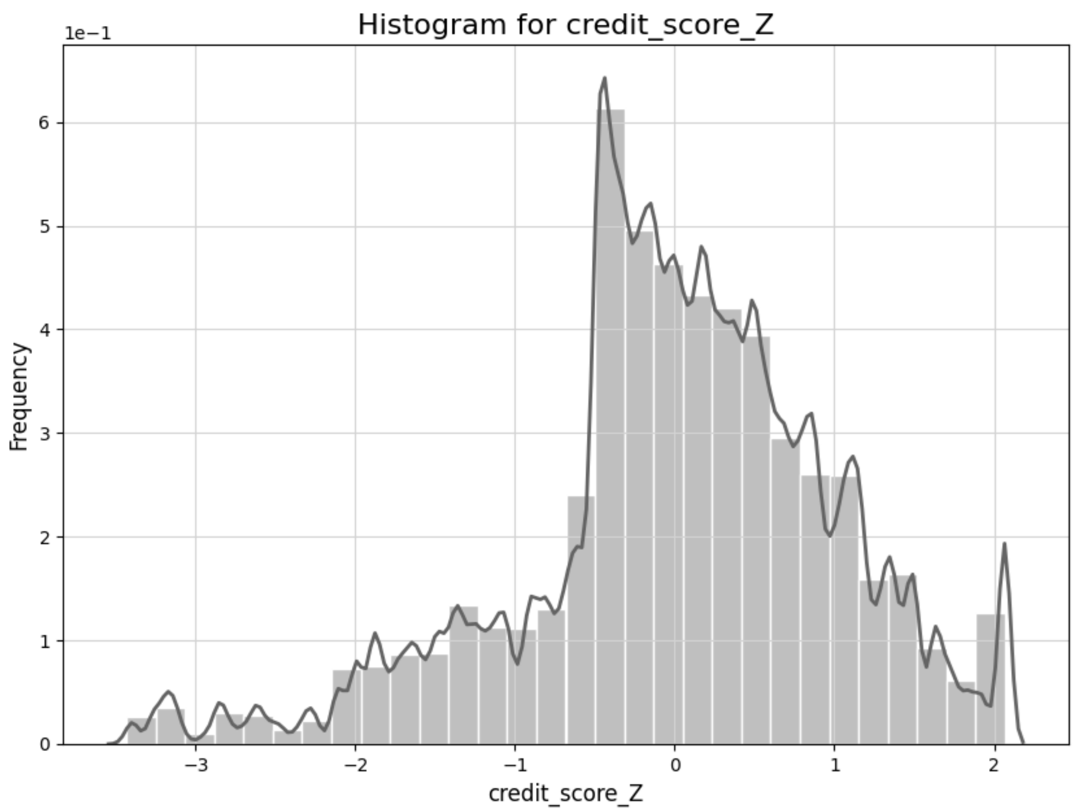


Original Variable	Logarithmic Variables	Scaled Variable	Reason
amount	log_amount	log_amount_Z	log transformation
yearly_income	log_yearly_income	log_yearly_income_Z	
per_capita_income	log_per_capita_income	log_per_capita_income_Z	
credit_limit	log_credit_limit	log_credit_limit_Z	
total_debt	log_total_debt	log_total_debt_Z	



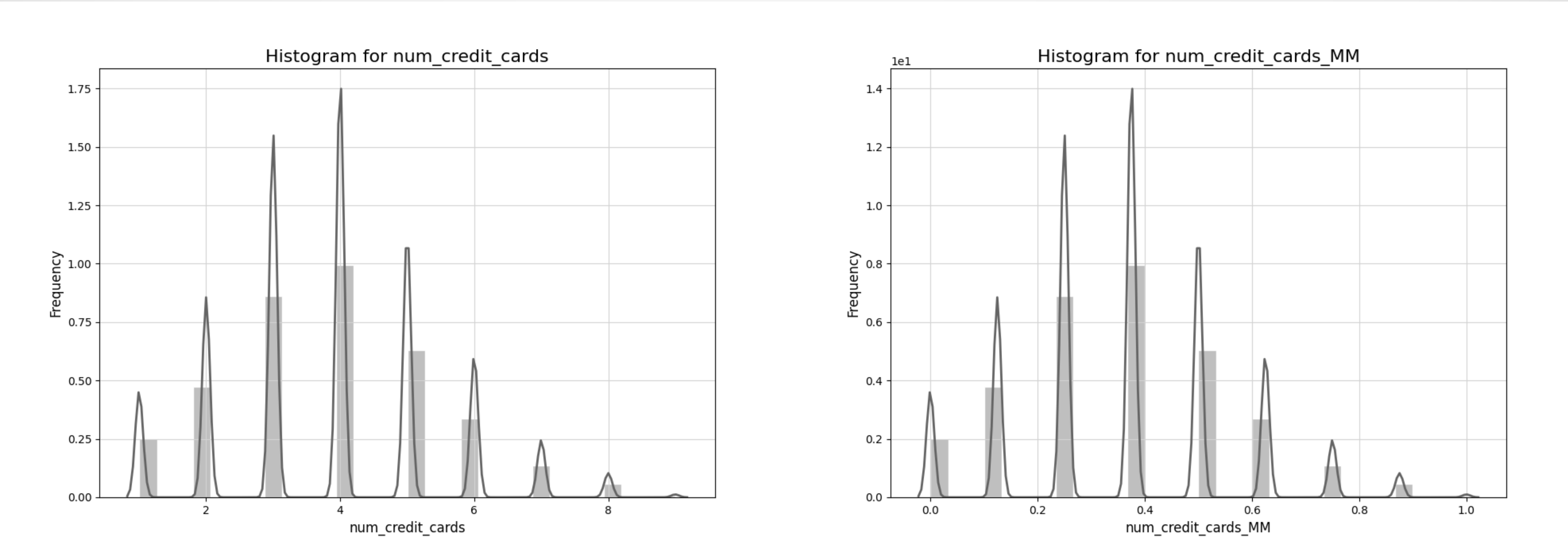
Scaling–Z score





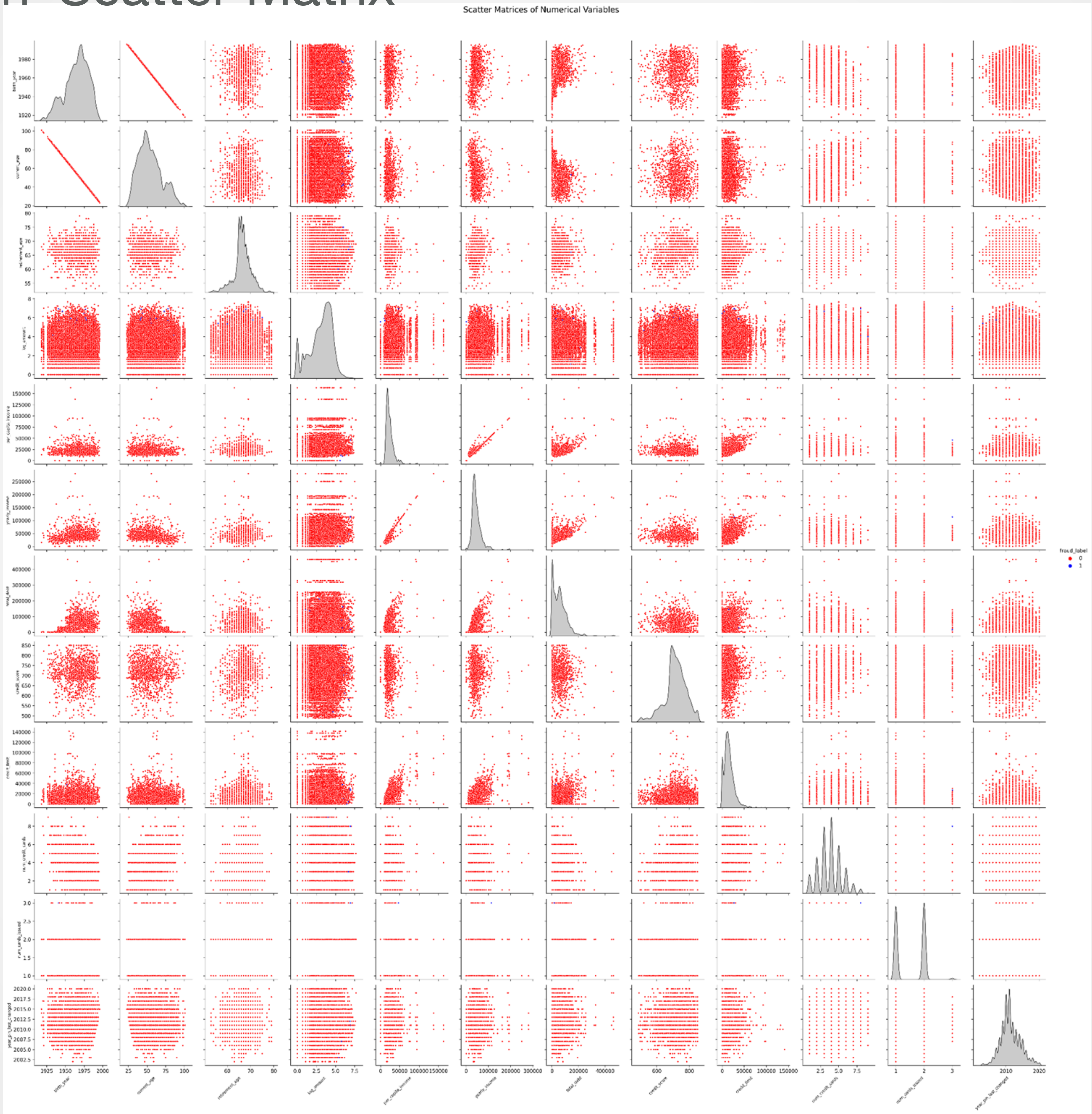
Variable	Scaled Variable	Reason
credit_score	credit_score_Z	highly bounded (Min 488.00, Max 850.00), close to normal, slightly left-skewed
current_age	current_age_Z	continuous, bounded variables, multimodal
retirement_age	retirement_age_Z	

Scaling–Min-Max Scaling

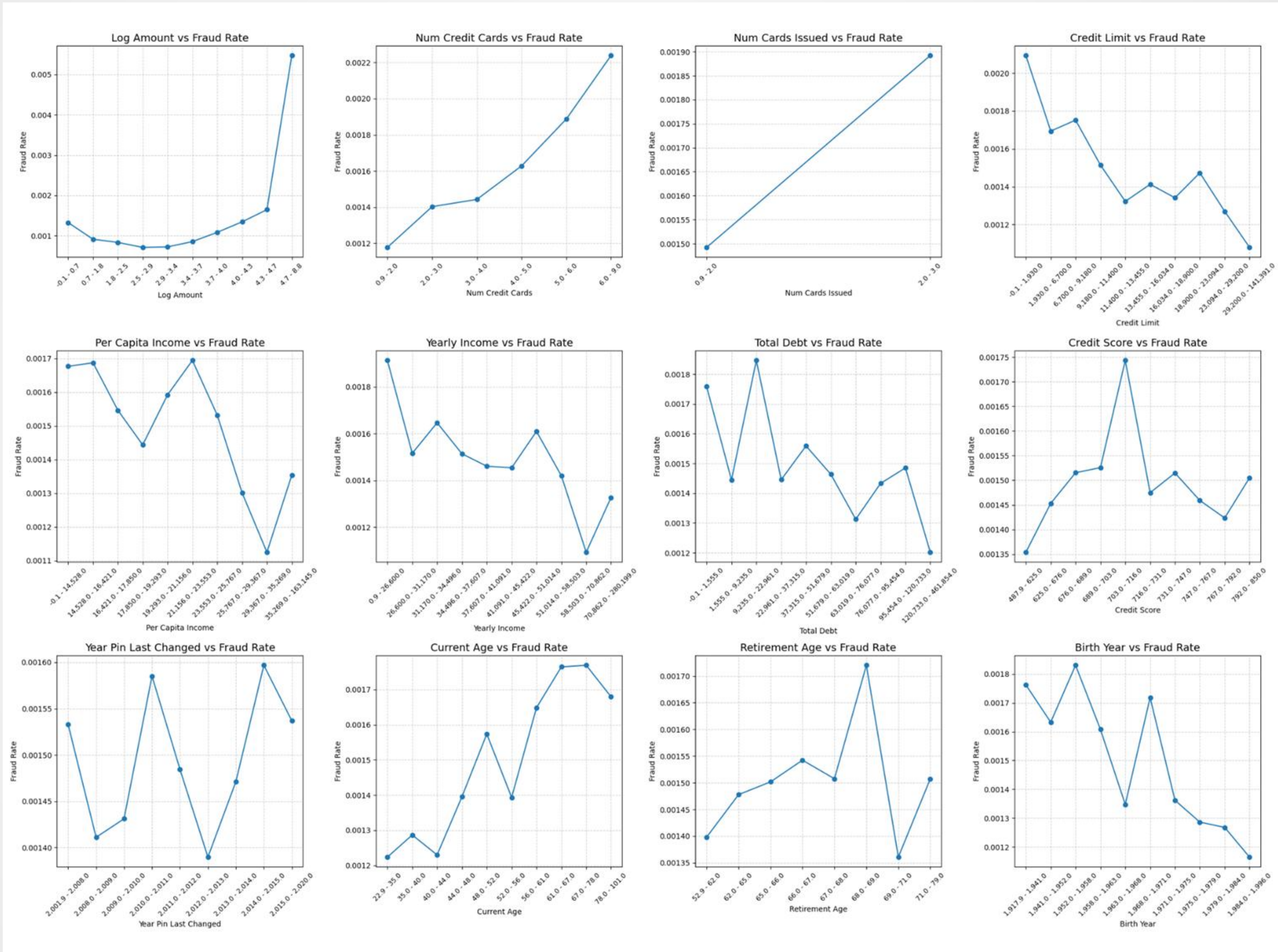


Variable	Reason
num_credit_cards	ordinal, discrete, limited ranged (Min 1.00, Max 9.00)
num_cards_issued	extremely tightly bounded (Min 1.00, Max 3.00)

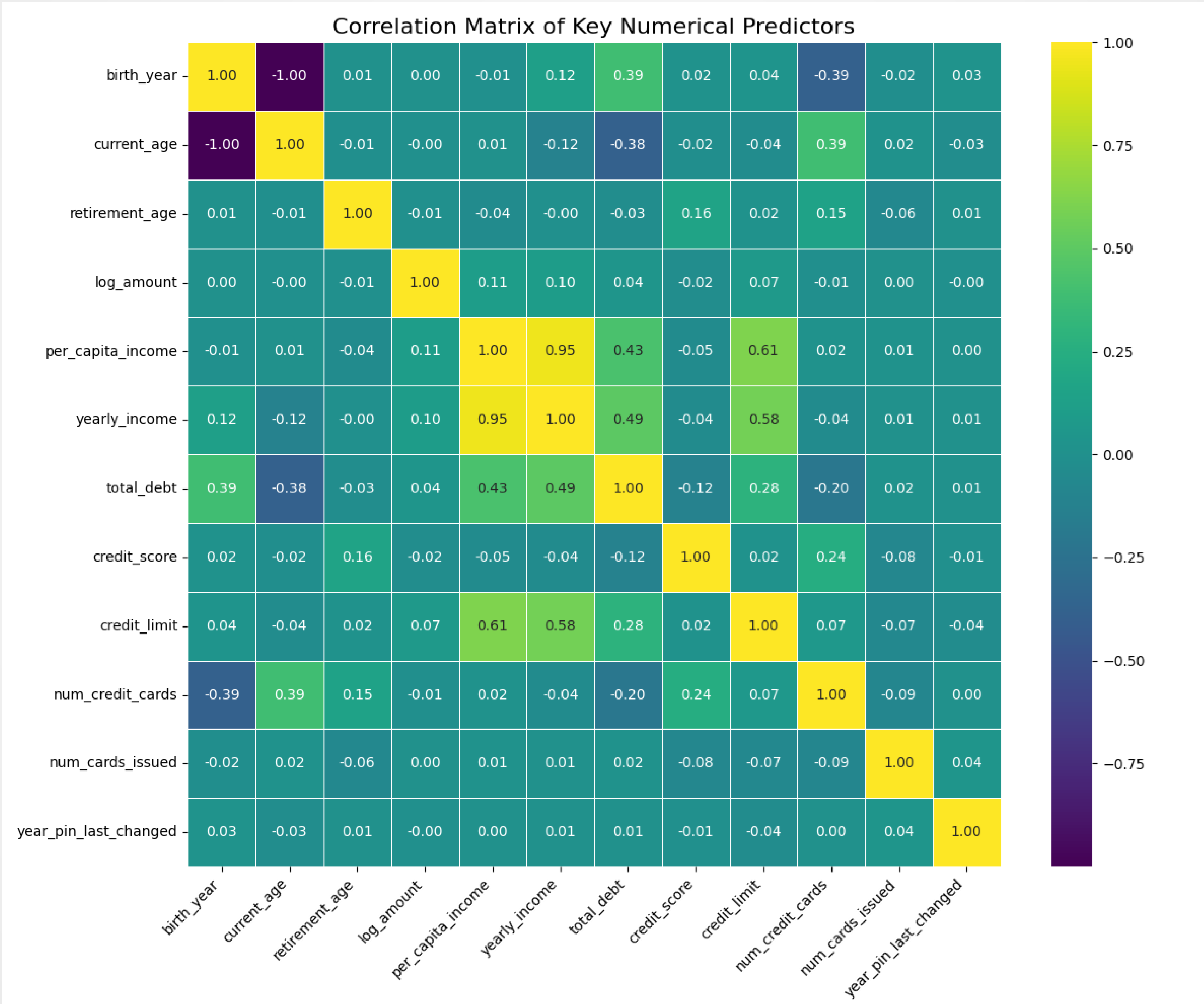
Correlation–Scatter Matrix



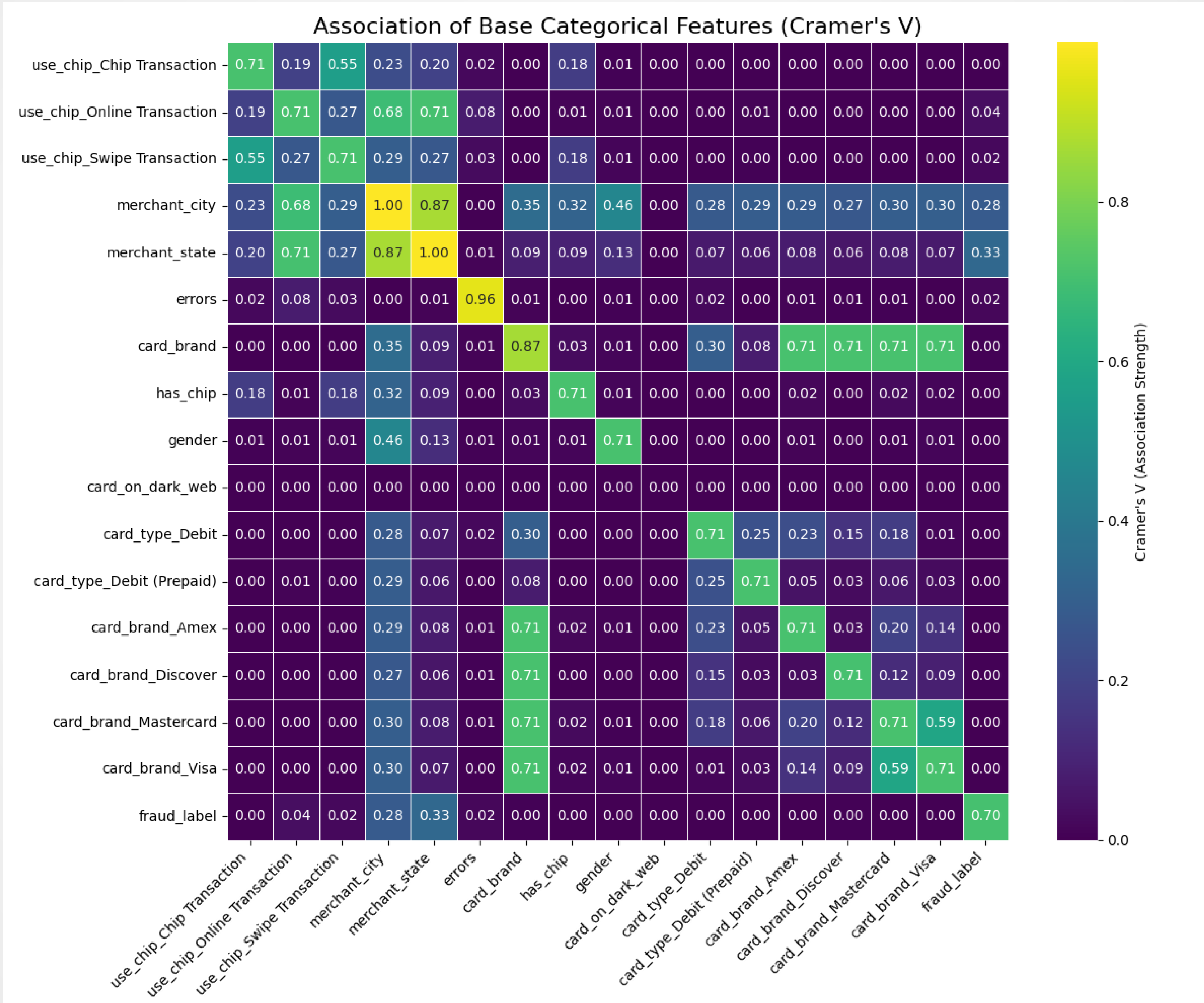
Correlation–Key Finding in Scatter Matrix



Correlation–Scatter Matrix



Correlation–Scatter Matrix



Benchmark Model—Logistic Regression

Linearity of logit assumption– check Box Tidwell Test

Index	Variables	p-value	Result
X_5	amount	0.00E+00	Violated
X_{10}	zip	3.95E-197	Violated
X_{11}	mcc_code	4.86E-155	Violated
X_1	transaction_id	2.69E-49	Violated
X_7	merchant_id	7.49E-13	Violated
X_{20}	current_age	8.55E-11	Violated
X_{21}	retirement_age	3.01E-09	Violated
X_{35}	card_number	5.35E-05	Violated
X_{40}	credit_limit	5.78E-02	Satisfied
X_{31}	credit_score	6.75E-02	Satisfied
X_{32}	num_credit_cards	7.52E-02	Satisfied

Benchmark Model—Logistic Regression

Multicollinearity assumption– check VIF index

Index	Variables	VIF Factor
X_{26}	card_number	22.243090
X_{29}	card_brand_Amex	4.419995
X_{38}	zip	2.302851
X_{27}	longitude	2.117485
X_{28}	per_capita_income	1.843738
X_{16}	use_chip_Chip Transaction	1.749374
X_{18}	use_chip_Swipe Transaction	1.718199
X_{29}	yearly_income	1.631605
X_{30}	total_debt	1.538366
X_{14}	zip_missing_flag	1.466621

Index	Variables	VIF Factor
X_{38}	zip	2.53292
X_{27}	longitude	2.460976
X_{18}	use_chip_Swipe Transaction	1.979256
X_{1}	transaction_id	1.768991
X_{40}	credit_limit	1.707588
X_{29}	yearly_income	1.627438
X_{30}	total_debt	1.592191
X_{20}	current_age	1.515529
X_{45}	card_type_Debit	1.414488
X_{39}	num_credit_cards	1.322601

Feature Selection Model—Logistic Regression with Stepwise

Stepwise completed until variables X_{49} card_brand_Mastercard

$$\log\left(\frac{P(Y=1)}{1-P(Y=1)}\right) = -4.6829 + 0.3586X_5 - 2.1209X_{18} - 7.17\times10^{-8}X_1$$
$$- 6.43\times10^{-5}X_{10} - 2.6983\times10^{-2}X_{27} - 4.2024\times10^{-3}X_{40}$$
$$+ 9.74\times10^{-6}X_7 + 0.1061X_{32} - 0.7411X_{15}$$
$$- 0.0297X_{29} - 1.50\times10^{-4}X_{11} - 0.3366X_{38}$$
$$- 0.0167X_{26} - 1.16\times10^{-4}X_3 + 7.44\times10^{-4}X_{31}$$
$$+ 0.0108X_{21} + 0.0514X_{49}$$

Train Accuracy: 0.9985, Test Accuracy: 0.9985

=== Overall Model Fit ===

	Measure	Value	Change_from_Base	Change_pvalue
0	-2 Log Likelihood (-2LL) value	131399.390	27711.222	0.0
1	Cox and Snell R2	0.004		
2	Nagelkerke R2	0.176		
3	Pseudo R2 (McFadden)	0.174		
4	Hosmer-Lemeshow χ2	27711.222		

Variables for Logistic Regression

Summary of variables

Index	Variables	Type	Index	Variables	Type
Y	Fraud_label	numeric	X_{21}	retirement_age	numeric
X_1	transaction_id	numeric	X_{26}	latitude	numeric
X_3	client_id	numeric	X_{27}	longitude	numeric
X_5	amount	numeric	X_{29}	yearly_income	numeric
X_7	merchant_id	numeric	X_{31}	credit_score	numeric
X_{10}	zip	numeric	X_{32}	num_credit_cards	numeric
X_{11}	mcc_code	numeric	X_{38}	has_chip	category
X_{15}	errors_missing_flag	category	X_{40}	credit_limit	numeric
X_{18}	use_chip_Swipe Transaction	category	X_{49}	card_brand_Mastercard	category